NYU Tandon School of Engineering

Algorithmic Portfolio Management

FRE7241 Homework 1

Portfolio Allocation and ETF Alpha

Methods, results, and interpretation

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What this report contains. Solutions to all eight questions, the historical bootstrap and optimization methods, complete annual alpha and ranking matrices, both requested plots, all annual CAPM tables, the full submitted R code, the execution output, and independent verification. The approximately ten-year bootstrap selects a VTI weight of 0.4543241. The annual ETF analysis produces an 8×21 alpha matrix. All ten independent numerical and artifact checks pass.

Abstract

This homework studies two historical portfolio questions using the supplied `rutils::etfenv` data. First, a paired historical block bootstrap constructs 1,000 cumulative VTI and IEF log-return scenarios at each investment horizon. A bounded scalar optimization maximizes the mean-to-standard-deviation ratio of the assignment's weighted cumulative log-return measure. The approximately ten-year optimal stock allocation is 45.43%, while allocations across the five requested horizons range from 35.97% to 45.88% and are not monotonic. Second, yearly endpoint intervals are used to estimate CAPM-style statistics for eight ETFs against VTI. The report presents the full annualized alpha and ranking matrices and a 2008–2009 alpha scatterplot. It distinguishes the package's annualization convention from realized returns, documents shared interval boundaries and partial years, and records the robust-regression warning. Independent moment-based optimization and ordinary least-squares calculations verify the numerical results.

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1 Data, conventions, and reproducibility

The solution follows the course's bootstrap, R functional-programming, and CAPM examples [1, 2, 3, 4]. The executable submission is `Panagiotis_Housos_hw1.R`. Its values are computed from the data; the professor's example output is used as a diagnostic, rather than inserted into the calculations.

Part I selects VTI and IEF prices, removes rows with missing prices, extracts a numeric matrix, and takes logarithms. The resulting paired sample contains 5962 observations from `2002-07-26` through `2026-04-07`. Part II selects VTI, IEF, VNQ, USO, XLY, XLP, XLE, XLF, and XLK, applies the assigned `na.locf()` and `na.omit()` preprocessing, and contains 5028 daily observations from `2006-04-11` through `2026-04-07`.

The supplied return series contains *log returns*, as confirmed against differences of log prices. This matters for interpreting both the portfolio metric and the regression outputs.

Two indexing conventions follow the assignment and lecture examples:

- A holding block of h price observations starts at row s and ends at row $s + h - 1$, spanning $h - 1$ daily return intervals. The code samples from all $N - h + 1$ valid starting rows, including the last possible block.
- A CAPM interval includes both neighboring endpoint rows. Consequently, adjacent intervals share one boundary observation. These are the assignment's endpoint windows, rather than strictly disjoint calendar-year samples.

All simulations use the specified seed 1121, the Mersenne–Twister generator, and rejection sampling. The repository records package versions, GitHub source commits, the R session, input-data fingerprints, complete scenario exports, and verification results. The analysis is historical and in sample; it does not evaluate an allocation or ranking strategy on a held-out period.

2 Part I: optimal stock and bond allocations

2.1 I.1: bootstrap scenarios and portfolio wealth measure

Let $L_{t,a} = \log P_{t,a}$ be asset a's log price. For bootstrap draw $b = 1, \ldots, B$, sample a starting index independently with replacement:

$$s_b \sim \text{Uniform}\{1, \ldots, N - h + 1\}, \qquad B = 1000. \tag{1}$$

The paired cumulative log-return scenario is

$$R^{(h)}_{a,b} = L_{s_b+h-1,a} - L_{s_b,a}. \tag{2}$$

The same historical dates are used for both assets, preserving their joint holding-period behavior. The transposed differences form a 2×1000 matrix, with assets in rows and scenarios in columns. For the initial approximately ten-year calculation, $h = 10 \times 252 = 2520$ price observations.

Table 1: The first five paired approximately ten-year bootstrap scenarios. Entries are cumulative log returns.

ETF	1	2	3	4	5
VTI	1.38891556	1.14632894	0.73140357	1.50944354	1.11841720
IEF	0.43275276	0.10217090	0.51986895	0.27570544	0.08607574

For stock weight w, the assignment defines

$$W_b(w) = wR^{(h)}_{\mathrm{VTI},b} + (1-w)R^{(h)}_{\mathrm{IEF},b}. \tag{3}$$

The implementation calculates this with a matrix product and uses `drop()` to obtain a length-1000 vector. Table 2 shows the first five values at $w = 0.5$.

Table 2: The first five portfolio-metric values at the 50/50 allocation, using the scenarios in Table 1.

	Draw 1	Draw 2	Draw 3	Draw 4	Draw 5
Equal weight	0.91083416	0.62424992	0.62563626	0.89257449	0.60224647

Interpretation. Although the assignment calls W "wealth," it is a weighted cumulative log-return metric. If the initial dollar allocation were held without rebalancing, terminal gross wealth per initial dollar would be

$$G_b(w) = w\exp(R^{(h)}_{\mathrm{VTI},b}) + (1-w)\exp(R^{(h)}_{\mathrm{IEF},b}), \tag{4}$$

and $\log G_b(w)$ generally differs from $W_b(w)$. The submitted solution optimizes the measure requested in the assignment.

2.2 I.2: risk adjustment and bounded optimization

For a given scenario matrix, define

$$J_h(w) = \frac{\overline{W(w)}}{s_{W(w)}}, \qquad w^\star_h = \arg\min_{0\le w\le 1}\{-J_h(w)\}. \tag{5}$$

The function `riskretfun(retm, weightv)` returns the negative ratio; `optimize()` minimizes it over `c(0,1)`. The bounds represent a fully invested, long-only allocation with weights w and $1-w$.

For the 50/50 approximately ten-year sample, the portfolio metric has mean 0.7109867 and sample standard deviation 0.1025262. Its mean/SD ratio is 6.934682, so the objective is its negative. Optimization gives

$$\boxed{w^\star_{\mathrm{VTI}} = 0.4543241, \qquad w^\star_{\mathrm{IEF}} = 1 - 0.4543241.} \tag{6}$$

Thus the optimized allocation is approximately 45.43% VTI and 54.57% IEF, with optimized ratio 7.043107.

This ratio describes the cumulative horizon metric across sampled windows. It is not an annualized daily Sharpe ratio and does not subtract a risk-free return. Independent inverse-covariance calculations agree with the numerical optimum within the optimizer's default tolerance.

2.3 I.3: allocations at the five requested horizons

The requested horizons are generated from the sample size:

$$\boldsymbol{h} = \text{round}(N(0.1, 0.2, 0.3, 0.4, 0.5)) = (596, 1192, 1789, 2385, 2981). \tag{7}$$

A named list stores one 2×1000 paired return matrix per horizon. An `sapply()` loop calls `optimize()` separately for each matrix and returns a named vector of optimal VTI allocations.

Table 3: Optimized allocations and portfolio-metric summary statistics across the five holding blocks. Trading years use $(h-1)/252$; metrics are cumulative log-return quantities.

Block obs.	Years	VTI	Mean	SD	Min	Max	Mean/SD
596	2.361	0.412702	0.144572	0.094930	-0.136319	0.374404	1.5229
1192	4.726	0.359727	0.268438	0.087073	0.069619	0.514457	3.0829
1789	7.095	0.416078	0.449749	0.076041	0.259623	0.724239	5.9146
2385	9.460	0.458836	0.634872	0.090716	0.444569	0.851139	6.9985
2981	11.825	0.432266	0.780490	0.099877	0.623356	1.128930	7.8145

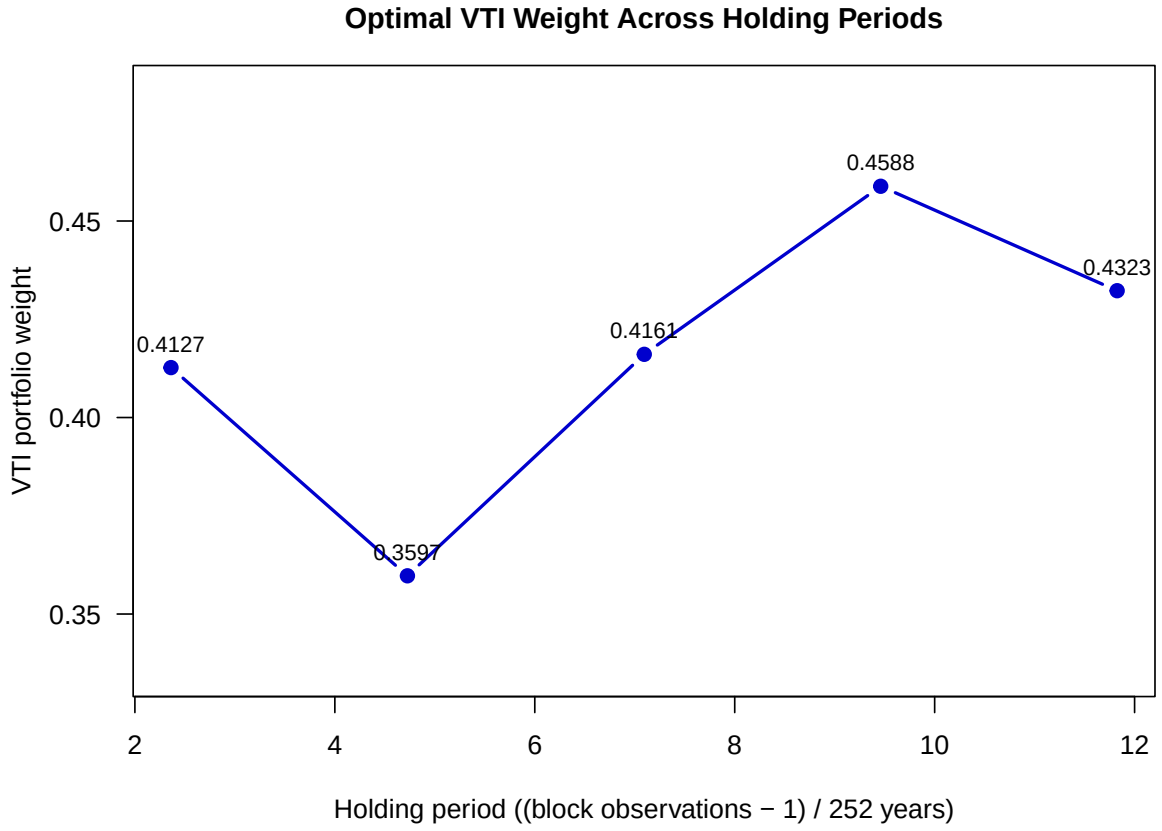


Figure 1: Optimal VTI allocations across the five requested holding blocks. The connecting line describes these five historical estimates.

Commentary. The stock weights are not monotonic in the holding period. They fall from 41.27% to 35.97%, rise to 45.88%, then fall to 43.23% at the longest horizon. Therefore the results do not support an unqualified claim that optimal stock exposure always increases with the investment horizon. The asset-return correlation is least negative in the shortest-horizon sample, but this descriptive pattern alone does not establish why an optimal weight changes.

The portfolio variance is

$$\mathrm{Var}(W) = w^2\sigma_V^2 + (1-w)^2\sigma_I^2 + 2w(1-w)\,\mathrm{Cov}(R_V, R_I). \tag{8}$$

Both expected cumulative returns and their joint dispersion change with the horizon, so the allocation depends on more than asset correlation.

2.4 I.4: portfolio distributions and historical interpretation

An `lapply()` loop over the named scenario list combines each matrix with its matching optimal weight. It produces five named length-1000 vectors. The first five shortest-horizon scenarios and portfolio values are recorded in Appendix A.

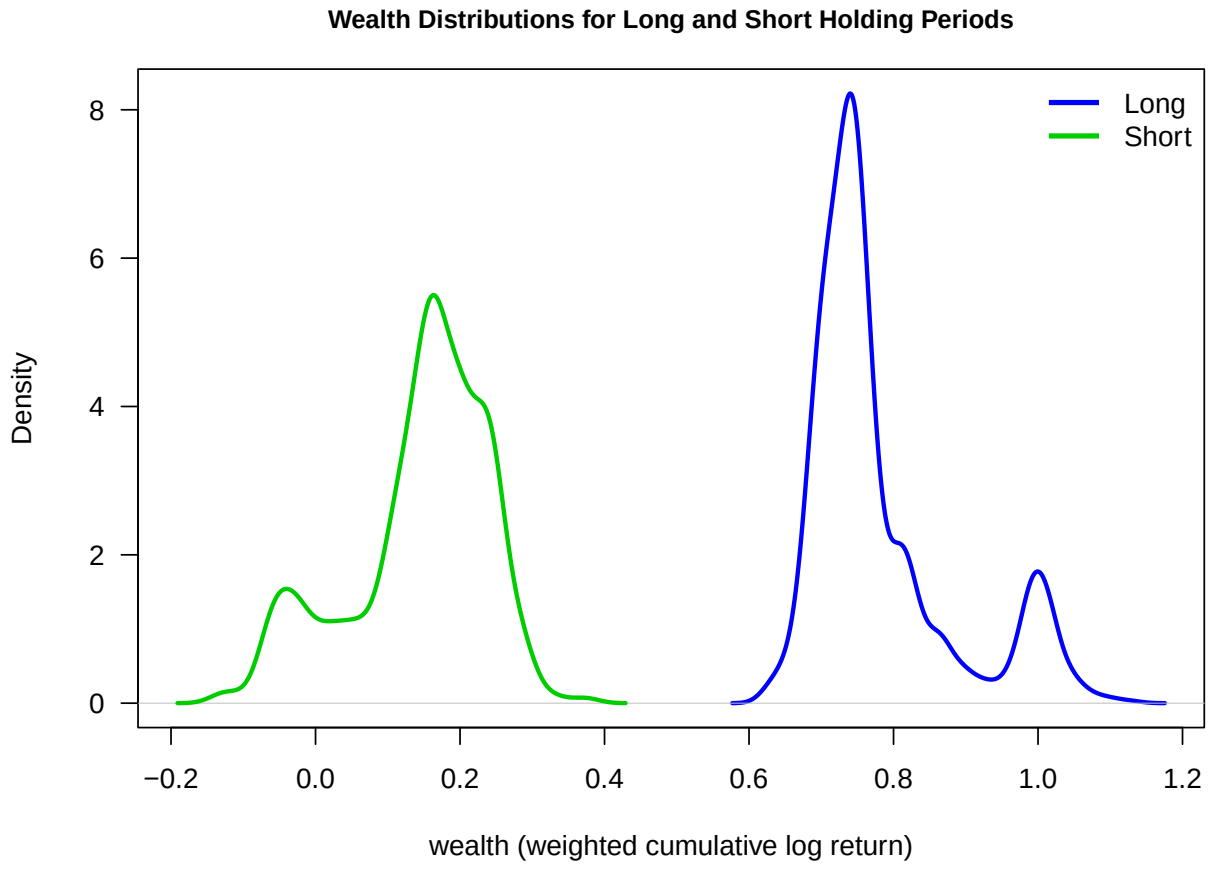


Figure 2: Kernel-density estimates of the optimized portfolio metric at the longest (blue) and shortest (green) holding blocks. The horizontal axis uses the assignment's cumulative log-return "wealth" convention.

The longest-horizon distribution lies farther to the right. Its sample mean is approximately 0.78049, compared with 0.14457 for the shortest horizon. The corresponding standard deviations are approximately 0.09988 and 0.09493. In particular, the rightward shift is not evidence that absolute dispersion

is smaller at the longest horizon. The shortest sample has 11.8% negative portfolio-metric values, while the longest sample has none.

These are frequencies among historical windows, not established future loss probabilities. Bootstrap draws are independent conditional on the fixed empirical dataset, but the underlying historical blocks overlap and may be sampled repeatedly. One thousand draws therefore do not represent one thousand newly observed independent market episodes. The overlap diagnostics in Appendix A quantify this distinction without treating them as an estimate of effective sample size. The favorable historical sample, changing market regimes, and the absence of an out-of-sample test limit any extrapolation of these optimized allocations.

3 Part II: annual ETF alpha and rankings

3.1 II.1: yearly endpoint intervals and CAPM tables

`rutils::calc_endpoints(retp, interval = "years")` supplies the year-end row positions, with initial zero and the final observation included. There are 21 intervals, labeled by their ending years 2006–2026. Following the assignment's example, interval i uses rows

$$\max(1, e_i), \ldots, e_{i+1}. \tag{9}$$

Thus the final interval begins on `2025-12-31`, includes the previous year's boundary observation, and ends on `2026-04-07`. The first interval has 183 observations and the last has 66 observations; the final interval contains 65 observations actually dated 2026. The first and last years are partial. All window dates and row counts appear in Appendix A.

For comparison asset a in a given interval, the ordinary regression is

$$r_{a,t} = \alpha_a + \beta_a r_{\mathrm{VTI},t} + \varepsilon_{a,t}, \tag{10}$$

where the supplied r are daily log returns and the assigned risk-free rate is zero. `table.CAPM()` compares the eight non-benchmark ETFs against VTI and calculates the requested summary table. Column names such as `IEF to VTI` are simplified to the ETF ticker, and the list is named with the interval-ending years.

Table 4: Complete CAPM table for the first interval, 2006. The remaining annual tables are in Appendix B. Values retain the package's four-decimal reporting convention.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0003	0.0007	-0.0016	0.0002	0.0004	-0.0004	0.0003	-0.0002
Beta	0.0326	0.9094	0.0865	0.9956	0.5466	1.2493	0.9259	1.1092
Alpha Robust	0.0003	0.0006	-0.0015	0.0003	0.0005	-0.0001	0.0002	-0.0002
Beta Robust	0.0523	0.9582	0.0881	0.9822	0.5670	1.2525	0.8963	1.1390
Beta+	0.0831	0.8206	0.6261	1.0077	0.3818	1.4792	0.8758	1.0898
Beta-	0.0188	0.6517	0.0731	0.9016	0.6239	1.5042	0.9472	0.9853
Beta+ Robust	0.0868	0.8887	0.6261	0.9673	0.3920	1.4617	0.8617	1.0265
Beta- Robust	0.0639	0.7227	0.0835	0.8966	0.6247	1.5130	0.9279	1.0427
R-squared	0.0069	0.4583	0.0015	0.7497	0.4770	0.3022	0.7175	0.7213
R-squared Robust	0.0183	0.4976	0.0015	0.7283	0.4658	0.3144	0.6502	0.7006
Annualized Alpha	0.0689	0.1926	-0.3242	0.0632	0.1090	-0.0886	0.0769	-0.0604
Correlation	0.0832	0.6770	0.0383	0.8658	0.6906	0.5498	0.8471	0.8493
Correlation p-value	0.2631	0.0000	0.6064	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1144	0.1089	0.2668	0.0631	0.0801	0.2099	0.0642	0.0765
Active Premium	-0.0577	0.1957	-0.4673	0.0684	0.0550	-0.0928	0.0739	-0.0578
Information Ratio	-0.5040	1.7976	-1.7512	1.0842	0.6870	-0.4421	1.1511	-0.7553
Treynor Ratio	2.2196	0.3582	-3.8978	0.1993	0.3385	0.0298	0.2203	0.0651

The package reports ordinary and robust coefficients, conditional up/down betas, correlations, and benchmark-relative performance measures. During the 2012 VNQ `Beta- Robust` calculation, a robust-

regression M-step reports nonconvergence. The submission retains this warning. It concerns that robust downside-beta estimate; the ordinary alpha/beta values and annualized alphas used below are independently verified and unaffected.

3.2 II.2: annualized alpha matrix

The code extracts each table's `Annualized Alpha` row with `unlist()` inside `sapply()`, obtaining an 8×21 matrix. Table 5 and Table 6 include every year and ETF.

Table 5: Annualized alpha, 2006–2016. Entries are decimal values; for example, 0.1205 corresponds to 12.05%.

ETF	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
IEF	0.0689	0.1137	0.1205	-0.0488	0.1229	0.1639	0.0790	-0.0470	0.1114	0.0152	0.0290
VNQ	0.1926	-0.2144	0.2848	-0.2019	0.0418	0.0730	0.0489	-0.2262	0.2165	0.0116	-0.0101
USO	-0.3242	0.4603	-0.4475	-0.0128	-0.1545	-0.0072	-0.2647	-0.1209	-0.4477	-0.4537	-0.0814
XLY	0.0632	-0.1814	0.0540	0.0706	0.0763	0.0481	0.0654	0.0602	-0.0286	0.1013	-0.0575
XLP	0.1090	0.0921	0.0858	0.0086	0.0341	0.1348	0.0166	0.0090	0.0800	0.0608	-0.0252
XLE	-0.0886	0.2754	0.0866	-0.0996	0.0129	0.0183	-0.1196	-0.0710	-0.1912	-0.2141	0.1115
XLF	0.0769	-0.2474	-0.0868	-0.3040	-0.0766	-0.1798	0.0701	-0.0409	0.0257	-0.0217	0.0622
XLK	-0.0604	0.1025	-0.1060	0.2115	-0.0430	0.0164	-0.0115	-0.0142	0.0523	0.0502	0.0123

Table 6: Annualized alpha, 2017–2026. The 2026 column is based on the partial final endpoint window.

ETF	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
IEF	0.0726	0.0061	0.1485	0.1170	-0.0249	-0.1376	0.0253	-0.0185	0.0827	-0.0016
VNQ	-0.0382	-0.0305	0.1478	-0.2243	0.1978	-0.1146	-0.1366	-0.0936	-0.0465	0.1760
USO	-0.0468	-0.1623	0.0594	-0.7307	0.3353	0.3262	-0.1133	0.1058	-0.1417	9.9429
XLY	0.0249	0.0743	-0.0200	0.0799	-0.0163	-0.1583	0.0398	-0.0272	-0.1046	-0.2319
XLP	0.0350	-0.0508	0.1040	-0.0376	0.0715	0.1145	-0.1125	0.0687	-0.0189	0.2001
XLE	-0.1553	-0.1372	-0.1606	-0.4797	0.1888	0.8886	-0.1496	-0.0440	-0.0361	2.0778
XLF	-0.0530	-0.0856	-0.0047	-0.2174	0.0957	0.0810	-0.1099	0.1168	0.0100	-0.2248
XLK	0.0696	0.0545	0.0496	0.1615	0.0111	-0.0488	0.1940	-0.1167	0.0062	-0.0094

Interpretation and units. Alpha is the fitted intercept relative to VTI exposure; it is not the ETF's realized total return. The requested `PerformanceAnalytics` table defines annualized alpha as

$$A_a = (1 + \widehat{\alpha}_{a,\text{daily}})^{252} - 1. \tag{11}$$

This is the package's convention, applied here to regressions of the supplied daily log returns. It differs from the annual log-intercept $252\widehat{\alpha}_{a,\text{daily}}$.

USO's 2026 entry of 9.9429 corresponds to approximately 994.29% *package-defined annualized alpha.* It extrapolates a short interval's estimated daily intercept, approximately 0.0095400, across 252 days. It does not mean that USO earned a realized 994.29% year-to-date return. The partial-year sample and nonlinear annualization make this an especially sensitive figure. More generally, a positive alpha can coexist with a negative realized return when the benchmark and fitted market exposure are unfavorable.

3.3 II.3: descending annual rankings

The final alpha column is sorted in descending order and its ticker names are extracted. The resulting order is

`USO, XLE, XLP, VNQ, IEF, XLK, XLF, XLY`.

An `apply()` loop repeats this operation across every year and produces the 8×21 ticker matrix shown below. Row 1 is the highest alpha within each interval, and row 8 is the lowest.

Table 7: Descending annual alpha rankings, 2006–2016.

Rank	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
1	VNQ	USO	VNQ	XLK	IEF	IEF	IEF	XLY	VNQ	XLY	XLE
2	XLP	XLE	IEF	XLY	XLY	XLP	XLF	XLP	IEF	XLP	XLF
3	XLF	IEF	XLE	XLP	VNQ	VNQ	XLY	XLK	XLP	XLK	IEF
4	IEF	XLK	XLP	USO	XLP	XLY	VNQ	XLF	XLK	IEF	XLK
5	XLY	XLP	XLY	IEF	XLE	XLE	XLP	IEF	XLF	VNQ	VNQ
6	XLK	XLY	XLF	XLE	XLK	XLK	XLK	XLE	XLY	XLF	XLP
7	XLE	VNQ	XLK	VNQ	XLF	USO	XLE	USO	XLE	XLE	XLY
8	USO	XLF	USO	XLF	USO	XLF	USO	VNQ	USO	USO	USO

Table 8: Descending annual alpha rankings, 2017–2026.

Rank	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
1	IEF	XLY	IEF	XLK	USO	XLE	XLK	XLF	IEF	USO
2	XLK	XLK	VNQ	IEF	VNQ	USO	XLY	USO	XLF	XLE
3	XLP	IEF	XLP	XLY	XLE	XLP	IEF	XLP	XLK	XLP
4	XLY	VNQ	USO	XLP	XLF	XLF	XLF	IEF	XLP	VNQ
5	VNQ	XLP	XLK	XLF	XLP	XLK	XLP	XLY	XLE	IEF
6	USO	XLF	XLF	VNQ	XLK	VNQ	USO	XLE	VNQ	XLK
7	XLF	XLE	XLY	XLE	XLY	IEF	VNQ	VNQ	XLY	XLF
8	XLE	USO	XLE	USO	IEF	XLY	XLE	XLK	USO	XLY

These are retrospective rankings of benchmark-adjusted regression alpha, rather than rankings of realized total returns. The leading ETF changes across years. The exercise does not implement or test a rule that uses a previous year's ranking to select the following year's portfolio.

3.4 II.4: 2008–2009 alpha scatterplot

The scatterplot places 2008 alpha on the horizontal axis and 2009 alpha on the vertical axis. Each point is labeled by ETF ticker; expanded axis limits provide space for the labels.

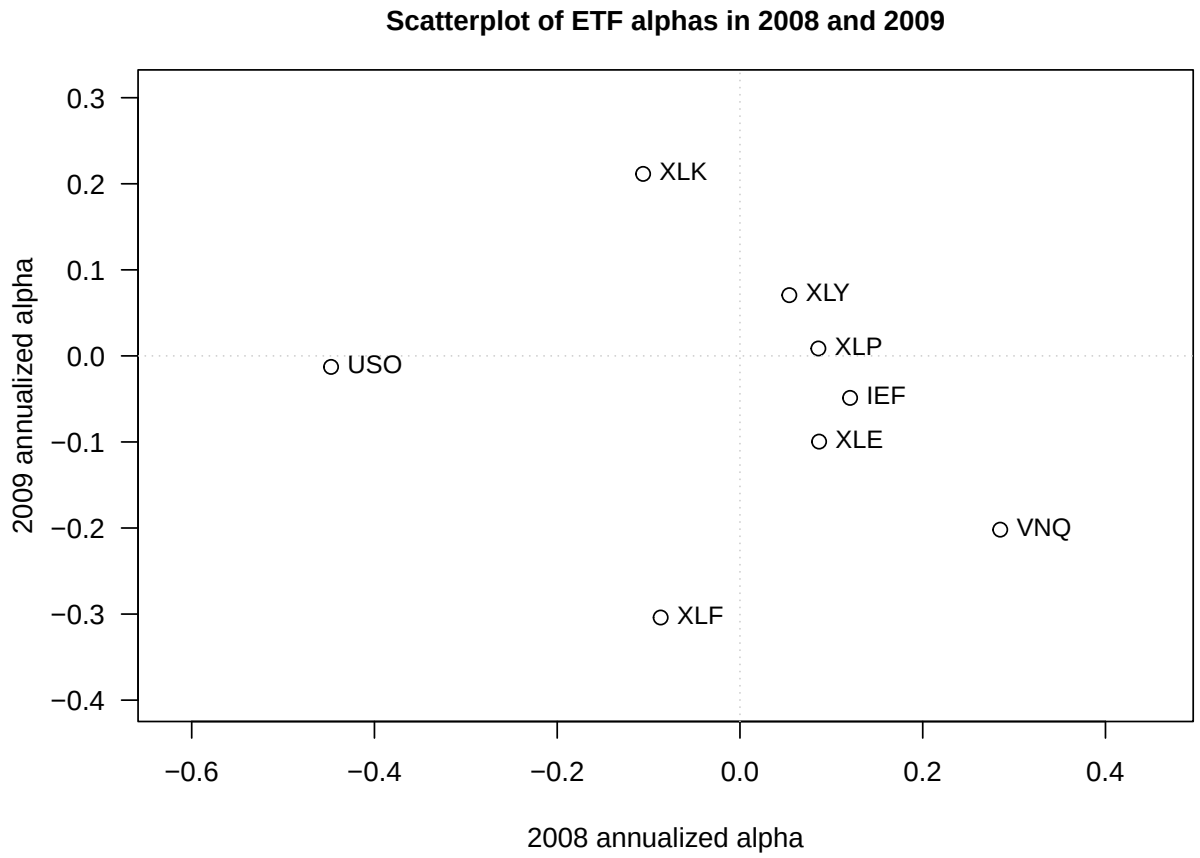


Figure 3: ETF annualized alpha in 2008 versus 2009, using the requested endpoint intervals and VTI benchmark.

Table 9: The eight pairs of annualized alphas used in Figure 3.

ETF	alpha_2008	alpha_2009
IEF	0.1205	-0.0488
VNQ	0.2848	-0.2019
USO	-0.4475	-0.0128
XLY	0.0540	0.0706
XLP	0.0858	0.0086
XLE	0.0866	-0.0996
XLF	-0.0868	-0.3040
XLK	-0.1060	0.2115

The cross-sectional Pearson correlation is -0.239298, indicating weak negative association among these eight points. VNQ changes from positive to negative alpha, while XLK changes from negative to positive. XLY and XLP remain positive; USO and XLF remain negative. Overall, 4 ETFs retain their alpha sign and 4 change sign.

The plot supplies little evidence of positive alpha persistence from 2008 to 2009 in this small ETF cross-section. Eight related assets over one pair of years cannot establish that alpha generally has no relationship across time. No predictive conclusion or statistical significance claim is inferred from this figure alone.

4 Verification and reproducible deliverables

The supplied sample output is a useful check, but numerical agreement alone does not establish a correct implementation. The repository's `hw1_results/verify_hw1.R` performs ten checks, including:

- reconstructing cumulative bootstrap returns by summing underlying daily log returns;
- comparing all horizon allocations with an independent moment-based optimum, using weights proportional to $\widehat{\Sigma}^{-1}\widehat{\mu}$ and checking the long-only bounds;
- testing the risk function on separate synthetic inputs;
- fitting ordinary regressions directly with `lm.fit()` to verify every year's intercept, beta, and annualized alpha;
- checking the annual ticker permutations, descending alpha order, matrix dimensions, and generated PNG artifacts.

All ten checks pass. The largest five-horizon allocation difference from the independent mathematical optimum is approximately 1.62×10^{-5}. The largest annualized-alpha difference before display rounding is approximately 4.98×10^{-5}, within the four-decimal table precision. The full verification record is reproduced in Appendix E.

The package includes the original assignment and reference images, the completed submission, this PDF and its LaTeX source, vector and raster figures, full scenario and annual-statistics exports, the console record, verification code, setup instructions, and environment/data manifests. The 1,000 individual scenario values are provided in machine-readable files alongside this report rather than printed as thousands of table rows.

From the repository root on Windows, run:

```
Rscript --vanilla setup.R
powershell -ExecutionPolicy Bypass -File build.ps1
```

The README gives equivalent R and LaTeX commands for other platforms. The setup installs missing packages and records the pinned course-package source commits; it does not silently replace existing installations.

References

[1] J. Pawlowski. *Univariate Investment Strategies*, FRE7241 course slides, January 20, 2026, especially slides 14–19.

[2] J. Pawlowski. *Functions*, FRE7241 course slides, January 20, 2026, especially slides 31–40.

[3] J. Pawlowski. *Date and Time Series Objects*, FRE7241 course slides, January 20, 2026, especially the xts indexing and endpoint sections.

[4] J. Pawlowski. *Data Management* and *Multivariate Time Series*, FRE7241 course slides, CAPM-table examples.

[5] `rutils` 0.2 and `PerformanceAnalytics` 2.1.0, installed package implementations and data. Exact GitHub source commits are recorded in `reproducibility/package-versions.csv`.

A Additional scenario and interval outputs

Table 10: First five paired scenarios at the shortest holding block, $h = 596$.

ETF	1	2	3	4	5
VTI	0.26369527	0.36955786	0.14826377	0.21959967	0.28329944
IEF	0.09298308	0.23527189	0.14635179	-0.20170717	0.21198804

Table 11: First five optimized portfolio-metric values at the shortest holding block.

	Draw 1	Draw 2	Draw 3	Draw 4	Draw 5
Optimal	0.16343630	0.29069195	0.14714087	-0.02783308	0.24141838

Table 12: Historical-window overlap diagnostics. These describe reuse of historical return intervals, not an effective sample size estimate.

Block obs.	Possible starts	Distinct draws	Overlap (percent)	Mean shared (percent)	Disjoint maximum
596	5367	905	21.036	10.747	10
1192	4771	899	43.685	22.875	5
1789	4174	886	67.432	36.798	3
2385	3578	883	89.489	52.412	2
2981	2982	840	100.000	66.909	2

Table 13: All CAPM endpoint intervals, including their shared boundaries and observation counts.

Year	start_date	end_date	observations	observations_in_named_year
2006	2006-04-11	2006-12-29	183	183
2007	2006-12-29	2007-12-31	252	251
2008	2007-12-31	2008-12-31	254	253
2009	2008-12-31	2009-12-31	253	252
2010	2009-12-31	2010-12-31	253	252
2011	2010-12-31	2011-12-30	253	252
2012	2011-12-30	2012-12-31	251	250
2013	2012-12-31	2013-12-31	253	252
2014	2013-12-31	2014-12-31	253	252
2015	2014-12-31	2015-12-31	253	252
2016	2015-12-31	2016-12-30	253	252
2017	2016-12-30	2017-12-29	252	251
2018	2017-12-29	2018-12-31	252	251
2019	2018-12-31	2019-12-31	253	252
2020	2019-12-31	2020-12-31	254	253
2021	2020-12-31	2021-12-31	253	252
2022	2021-12-31	2022-12-30	252	251
2023	2022-12-30	2023-12-29	251	250
2024	2023-12-29	2024-12-31	253	252
2025	2024-12-31	2025-12-31	251	250
2026	2025-12-31	2026-04-07	66	65

Table 14: First six rows of the final endpoint interval. Values are daily log returns.

Date	VTI	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
2025-12-31	-0.00766583	-0.00332226	-0.00821568	-0.00835138	-0.00792430	-0.00603223	-0.00557602	-0.00745797	-0.00995239
2026-01-02	0.00309718	-0.00083229	0.00033896	-0.00289603	-0.00891661	0.00012872	0.02080642	0.00291705	0.00228952
2026-01-05	0.00749467	0.00301377	0.00225683	0.01810654	0.01634229	-0.00438597	0.02680083	0.02161078	0.00221515
2026-01-06	0.00697010	-0.00072663	0.00729971	-0.02465345	0.00638027	0.00155019	-0.02701991	0.00479873	0.01393918
2026-01-07	-0.00340546	0.00186741	-0.00978250	-0.01056503	-0.00140513	-0.01103263	-0.01123731	-0.01410612	-0.00081861
2026-01-08	0.00026463	-0.00301033	0.00888695	0.03976523	0.01656998	0.02347586	0.03097979	0.00520134	-0.01575161

Table 15: Last six rows of the final endpoint interval. Values are daily log returns.

Date	VTI	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
2026-03-30	-0.00451367	0.00705749	0.00378593	0.04433273	-0.00018927	0.00122205	-0.00963708	0.01143820	-0.01880251
2026-03-31	0.02887203	0.00178281	0.01556584	-0.02007225	0.03093799	0.00122055	-0.01136191	0.02066993	0.04148060
2026-04-01	0.00754603	-0.00086829	0.00360117	-0.02514655	0.00749615	-0.00636321	-0.03809826	0.00141686	0.01501092
2026-04-02	0.00160742	0.00231214	0.01350090	0.10566670	-0.01514138	0.00526478	0.00473694	0.00181873	0.00797346
2026-04-06	0.00446863	-0.00262785	0.00143973	0.00736838	0.00819563	0.00935893	0.00723118	0.00704157	0.00579244
2026-04-07	0.00067626	0.00252286	0.00143766	-0.00620896	-0.01171546	-0.01708192	0.00801072	0.00000000	0.00474090

B Complete annual CAPM tables

The first interval's complete table appears in Table 4. The following tables give all remaining intervals. The reported 2012 VNQ robust downside-beta estimate is subject to the convergence warning described in the main text. All ordinary-regression outputs used for alpha extraction were independently verified.

Table 16: Complete CAPM statistics for 2007.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0004	-0.0010	0.0015	-0.0008	0.0003	0.0010	-0.0011	0.0004
Beta	-0.1955	1.3558	0.2265	0.9905	0.5429	1.2519	1.3505	0.8901
Alpha Robust	0.0006	-0.0015	0.0015	-0.0008	0.0001	0.0013	-0.0013	0.0006
Beta Robust	-0.2015	1.4118	0.2271	0.9975	0.5321	1.2311	1.2200	0.8941
Beta+	-0.1641	1.5105	-0.0286	1.0324	0.5197	1.1275	1.5990	0.8571
Beta-	-0.2338	1.1428	0.5076	0.9495	0.5651	1.3866	1.3933	0.8877
Beta+ Robust	-0.1642	1.5302	-0.0496	1.0757	0.4476	1.1674	1.3735	0.8546
Beta- Robust	-0.2497	1.1677	0.5645	0.9637	0.5941	1.3873	1.2273	0.9514
R-squared	0.2685	0.6110	0.0161	0.8081	0.6314	0.6253	0.7896	0.7393
R-squared Robust	0.2815	0.5838	0.0150	0.7066	0.5536	0.5966	0.6567	0.7028
Annualized Alpha	0.1137	-0.2144	0.4603	-0.1814	0.0921	0.2754	-0.2474	0.1025
Correlation	-0.5182	0.7817	0.1269	0.8989	0.7946	0.7908	0.8886	0.8598
Correlation p-value	0.0000	0.0000	0.0442	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1952	0.1795	0.3046	0.0761	0.0973	0.1578	0.1230	0.0851
Active Premium	0.0647	-0.2285	0.3829	-0.1906	0.0781	0.2768	-0.2559	0.0992
Information Ratio	0.3314	-1.2726	1.2571	-2.5044	0.8027	1.7535	-2.0801	1.1652
Treynor Ratio	-0.5178	-0.1415	1.8517	-0.1555	0.2112	0.2503	-0.1624	0.1525

Table 17: Complete CAPM statistics for 2008.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0005	0.0010	-0.0024	0.0002	0.0003	0.0003	-0.0004	-0.0004
Beta	-0.1171	1.5139	0.4982	0.9785	0.5435	1.2646	1.5142	0.9235
Alpha Robust	0.0005	0.0018	-0.0035	0.0008	0.0007	0.0006	-0.0018	-0.0003
Beta Robust	-0.1391	1.4130	0.5157	1.0993	0.5312	1.2976	1.6308	0.8998
Beta+	-0.0526	1.3544	0.3942	0.8736	0.5775	1.3027	1.3227	0.9885
Beta-	-0.0897	1.5381	0.8435	0.9855	0.5353	1.4673	1.5151	0.8389
Beta+ Robust	-0.0695	1.6720	0.3519	1.2150	0.3946	1.3258	2.0654	0.9100
Beta- Robust	-0.1105	1.2712	0.9667	1.0252	0.5458	1.5159	1.6221	0.8075
R-squared	0.2279	0.6868	0.1500	0.7755	0.6920	0.6745	0.7445	0.8554
R-squared Robust	0.2913	0.6081	0.1370	0.7305	0.5509	0.4766	0.7029	0.7611
Annualized Alpha	0.1205	0.2848	-0.4475	0.0540	0.0858	0.0866	-0.0868	-0.1060
Correlation	-0.4774	0.8287	0.3873	0.8806	0.8319	0.8213	0.8628	0.9249
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.4538	0.4564	0.5137	0.2102	0.2325	0.3660	0.4090	0.1545
Active Premium	0.5963	-0.0944	-0.1969	0.0259	0.2351	-0.0809	-0.2276	-0.0425
Information Ratio	1.3140	-0.2069	-0.3834	0.1233	1.0113	-0.2211	-0.5565	-0.2752
Treynor Ratio	-1.5136	-0.3391	-1.2362	-0.4017	-0.3383	-0.3953	-0.4270	-0.4997

Table 18: Complete CAPM statistics for 2009.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0002	-0.0009	-0.0001	0.0003	0.0000	-0.0004	-0.0014	0.0008
Beta	-0.1086	2.0219	1.0301	1.0826	0.4917	1.1690	2.0820	0.8736
Alpha Robust	-0.0003	-0.0011	-0.0005	-0.0001	0.0002	-0.0007	-0.0019	0.0008
Beta Robust	-0.1332	2.0292	1.0903	1.0406	0.5057	1.1778	1.9357	0.8782
Beta+	-0.1159	2.1454	0.8472	1.1195	0.4665	1.0409	2.3685	0.8611
Beta-	-0.1526	1.9212	1.3524	1.0254	0.5075	1.2769	2.0814	0.9146
Beta+ Robust	-0.1396	2.2680	1.0456	1.0266	0.5846	1.0342	2.1006	0.8820
Beta- Robust	-0.1841	1.9226	1.2644	1.0369	0.5005	1.2682	1.8659	0.9250
R-squared	0.0956	0.7245	0.3250	0.8723	0.5912	0.8028	0.7693	0.8418
R-squared Robust	0.1385	0.6593	0.3090	0.8079	0.5644	0.7351	0.5847	0.7650
Annualized Alpha	-0.0488	-0.2019	-0.0128	0.0706	0.0086	-0.0996	-0.3040	0.2115
Correlation	-0.3091	0.8512	0.5701	0.9339	0.7689	0.8960	0.8771	0.9175
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.3124	0.4350	0.4007	0.1140	0.1760	0.1628	0.4241	0.1077
Active Premium	-0.3391	-0.1430	-0.1055	0.1019	-0.1261	-0.1032	-0.2697	0.2205
Information Ratio	-1.0853	-0.3287	-0.2633	0.8936	-0.7162	-0.6336	-0.6360	2.0469
Treynor Ratio	0.7372	0.0574	0.1490	0.3334	0.2704	0.1333	-0.0051	0.5489

Table 19: Complete CAPM statistics for 2010.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0005	0.0002	-0.0007	0.0003	0.0001	0.0001	-0.0003	-0.0002
Beta	-0.2098	1.2692	1.0940	1.0579	0.5685	1.1684	1.2630	0.9511
Alpha Robust	0.0007	0.0005	-0.0007	0.0002	-0.0000	0.0000	-0.0006	-0.0001
Beta Robust	-0.2242	1.2886	1.0949	1.0590	0.5862	1.1299	1.2378	0.9308
Beta+	-0.2074	1.2656	1.1743	1.0620	0.5755	1.2594	1.2109	0.9270
Beta-	-0.2445	1.1746	1.0454	1.0499	0.6018	1.1061	1.2187	0.9615
Beta+ Robust	-0.2473	1.3038	1.1763	1.0686	0.5758	1.3337	1.2070	0.9205
Beta- Robust	-0.2331	1.1691	1.0301	1.0468	0.6268	1.0765	1.1982	0.9452
R-squared	0.2558	0.7561	0.5160	0.8968	0.7482	0.8510	0.8413	0.8825
R-squared Robust	0.2847	0.6946	0.5088	0.8323	0.6701	0.7688	0.7739	0.8107
Annualized Alpha	0.1229	0.0418	-0.1545	0.0763	0.0341	0.0129	-0.0766	-0.0430
Correlation	-0.5058	0.8695	0.7184	0.9470	0.8650	0.9225	0.9172	0.9394
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.2330	0.1421	0.1964	0.0671	0.1003	0.0955	0.1124	0.0647
Active Premium	-0.0575	0.0733	-0.1845	0.0926	-0.0244	0.0323	-0.0615	-0.0576
Information Ratio	-0.2467	0.5155	-0.9392	1.3800	-0.2429	0.3383	-0.5478	-0.8897
Treynor Ratio	-0.4047	0.1699	-0.0385	0.2221	0.2076	0.1495	0.0640	0.0891

Table 20: Complete CAPM statistics for 2011.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0006	0.0003	-0.0000	0.0002	0.0005	0.0001	-0.0008	0.0001
Beta	-0.2270	1.0864	0.7604	0.9627	0.5507	1.2214	1.3092	0.8987
Alpha Robust	0.0006	0.0006	-0.0007	0.0004	0.0006	-0.0000	-0.0008	0.0000
Beta Robust	-0.2424	1.0741	0.8380	0.9587	0.5465	1.2131	1.3022	0.9081
Beta+	-0.2031	1.1883	0.7073	0.9262	0.4958	1.2219	1.4157	0.9185
Beta-	-0.2192	1.1042	0.7981	0.9784	0.5951	1.1886	1.3260	0.8816
Beta+ Robust	-0.2622	1.0915	0.7488	0.9245	0.5082	1.2288	1.4041	0.9092
Beta- Robust	-0.2088	1.1884	0.9201	0.9632	0.5603	1.1822	1.4032	0.8879
R-squared	0.4426	0.8173	0.3064	0.9205	0.7982	0.8544	0.8876	0.9123
R-squared Robust	0.4666	0.7020	0.2938	0.8736	0.7439	0.7795	0.7982	0.8623
Annualized Alpha	0.1639	0.0730	-0.0072	0.0481	0.1348	0.0183	-0.1798	0.0164
Correlation	-0.6653	0.9041	0.5535	0.9594	0.8934	0.9243	0.9421	0.9551
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.3010	0.1251	0.2807	0.0685	0.1268	0.1323	0.1343	0.0712
Active Premium	0.1772	0.0591	-0.0340	0.0466	0.1477	-0.0014	-0.1954	0.0185
Information Ratio	0.5888	0.4722	-0.1212	0.6802	1.1649	-0.0105	-1.4551	0.2594
Treynor Ratio	-0.6944	0.0364	-0.0705	0.0281	0.2326	-0.0172	-0.1642	-0.0012

Table 21: Complete CAPM statistics for 2012. VNQ robust downside beta is subject to the recorded convergence warning.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0003	0.0002	-0.0012	0.0003	0.0001	-0.0005	0.0003	-0.0000
Beta	-0.2453	0.7477	1.1193	0.9635	0.5481	1.1982	1.1995	1.0199
Alpha Robust	0.0004	0.0004	-0.0016	0.0002	0.0002	-0.0006	0.0003	-0.0000
Beta Robust	-0.2503	0.7344	1.0664	0.9830	0.5464	1.2150	1.2013	1.0160
Beta+	-0.2197	0.7622	1.1724	0.9424	0.5803	1.1610	1.0862	1.1355
Beta-	-0.2505	0.8688	1.2800	1.0475	0.6113	1.1603	1.1982	1.0058
Beta+ Robust	-0.2257	0.7348	0.7236	0.9649	0.5745	1.1119	1.0380	1.0941
Beta- Robust	-0.2180	0.9346	1.4464	1.1398	0.6036	1.0779	1.1926	1.0592
R-squared	0.3387	0.5423	0.3399	0.8231	0.5922	0.7544	0.8186	0.8357
R-squared Robust	0.3637	0.4904	0.2896	0.7551	0.5250	0.7262	0.7797	0.7946
Annualized Alpha	0.0790	0.0489	-0.2647	0.0654	0.0166	-0.1196	0.0701	-0.0115
Correlation	-0.5820	0.7364	0.5830	0.9073	0.7695	0.8686	0.9048	0.9141
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1701	0.0964	0.2061	0.0590	0.0845	0.0938	0.0789	0.0596
Active Premium	-0.1111	0.0116	-0.3089	0.0672	-0.0521	-0.1153	0.1092	-0.0123
Information Ratio	-0.6530	0.1201	-1.4989	1.1378	-0.6165	-1.2302	1.3841	-0.2061
Treynor Ratio	-0.1583	0.2160	-0.1420	0.2253	0.1786	0.0289	0.2160	0.1350

Table 22: Complete CAPM statistics for 2013.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0002	-0.0010	-0.0005	0.0002	0.0000	-0.0003	-0.0002	-0.0001
Beta	-0.0644	0.9486	0.6397	1.0339	0.7750	1.0753	1.1706	0.8632
Alpha Robust	0.0000	-0.0006	-0.0009	0.0002	0.0002	-0.0005	-0.0002	-0.0001
Beta Robust	-0.0854	0.8629	0.7029	1.0368	0.7396	1.0801	1.1792	0.8399
Beta+	-0.1396	0.9115	0.6242	0.9696	0.8067	0.9819	1.1990	0.9722
Beta-	-0.0303	0.9877	0.8503	1.0179	0.6742	1.1737	1.0677	0.9602
Beta+ Robust	-0.2119	0.9281	0.6062	0.9659	0.8070	1.0162	1.2198	0.8370
Beta- Robust	-0.0964	0.7669	1.0690	1.0112	0.5191	1.2623	1.0670	0.9511
R-squared	0.0141	0.4912	0.1641	0.8577	0.6011	0.7194	0.8662	0.7152
R-squared Robust	0.0245	0.3944	0.1662	0.8428	0.5219	0.6933	0.8373	0.6825
Annualized Alpha	-0.0470	-0.2262	-0.1209	0.0602	0.0090	-0.0710	-0.0409	-0.0142
Correlation	-0.1189	0.7008	0.4051	0.9261	0.7753	0.8482	0.9307	0.8457
Correlation p-value	0.0590	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1358	0.1101	0.1694	0.0481	0.0763	0.0769	0.0559	0.0639
Active Premium	-0.4154	-0.3264	-0.2964	0.0937	-0.0781	-0.0716	0.0089	-0.0737
Information Ratio	-3.0584	-2.9645	-1.7492	1.9468	-1.0236	-0.9303	0.1597	-1.1528
Treynor Ratio	1.0455	0.0228	0.0807	0.4272	0.3483	0.2571	0.3049	0.3178

Table 23: Complete CAPM statistics for 2014.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0004	0.0008	-0.0024	-0.0001	0.0003	-0.0008	0.0001	0.0002
Beta	-0.1953	0.5164	0.3165	1.0132	0.5701	1.0802	0.9833	0.9808
Alpha Robust	0.0005	0.0010	-0.0009	-0.0002	0.0002	-0.0006	-0.0001	0.0003
Beta Robust	-0.2108	0.4917	0.2417	1.0461	0.6151	1.0255	0.9964	0.9823
Beta+	-0.1390	0.6322	0.1547	0.8376	0.7288	1.0735	0.9663	0.9598
Beta-	-0.2140	0.3824	0.3950	1.0811	0.5956	0.9731	1.0343	0.9385
Beta+ Robust	-0.1597	0.6170	0.2883	0.8929	0.6762	1.1146	1.0165	1.0115
Beta- Robust	-0.2321	0.3762	0.3341	1.1002	0.6313	1.0098	1.0521	0.9681
R-squared	0.2103	0.2928	0.0243	0.7703	0.3873	0.4540	0.8052	0.7277
R-squared Robust	0.2372	0.2660	0.0157	0.7817	0.4696	0.4519	0.7555	0.7491
Annualized Alpha	0.1114	0.2165	-0.4477	-0.0286	0.0800	-0.1912	0.0257	0.0523
Correlation	-0.4585	0.5411	0.1559	0.8777	0.6223	0.6738	0.8973	0.8530
Correlation p-value	0.0000	0.0000	0.0130	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1457	0.1089	0.2462	0.0643	0.0971	0.1380	0.0562	0.0698
Active Premium	-0.0374	0.1658	-0.5634	-0.0328	0.0294	-0.2152	0.0249	0.0533
Information Ratio	-0.2566	1.5230	-2.2885	-0.5094	0.3026	-1.5600	0.4430	0.7647
Treynor Ratio	-0.4302	0.5562	-1.3964	0.0875	0.2645	-0.0868	0.1488	0.1782

Table 24: Complete CAPM statistics for 2015.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0001	0.0000	-0.0024	0.0004	0.0002	-0.0010	-0.0001	0.0002
Beta	-0.1524	0.7478	0.9677	0.9935	0.7879	1.1792	1.0682	1.0749
Alpha Robust	0.0002	0.0009	-0.0033	0.0004	0.0003	-0.0012	-0.0003	0.0001
Beta Robust	-0.1794	0.6867	0.9685	0.9922	0.7971	1.1993	1.0517	1.0391
Beta+	-0.1267	0.8209	1.0385	0.9526	0.8089	1.2794	1.0795	1.0937
Beta-	-0.1684	0.7554	0.9363	1.0589	0.7441	1.1167	1.1376	1.0173
Beta+ Robust	-0.1392	0.7221	0.7785	0.9541	0.8263	1.2972	1.0693	1.0224
Beta- Robust	-0.1897	0.6291	1.0555	1.0496	0.7599	1.1761	1.1237	1.0096
R-squared	0.1249	0.4427	0.1239	0.8628	0.7563	0.5266	0.8613	0.8630
R-squared Robust	0.1810	0.4201	0.1278	0.8393	0.7312	0.4575	0.8289	0.8255
Annualized Alpha	0.0152	0.0116	-0.4537	0.1013	0.0608	-0.2141	-0.0217	0.0502
Correlation	-0.3534	0.6654	0.3520	0.9289	0.8696	0.7257	0.9281	0.9290
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1867	0.1339	0.3934	0.0606	0.0757	0.1731	0.0664	0.0665
Active Premium	0.0318	0.0099	-0.4850	0.0977	0.0634	-0.2258	-0.0254	0.0448
Information Ratio	0.1705	0.0741	-1.2329	1.6124	0.8371	-1.3041	-0.3822	0.6741
Treynor Ratio	-0.0917	-0.0106	-0.5196	0.0804	0.0578	-0.2066	-0.0404	0.0251

Table 25: Complete CAPM statistics for 2016.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0001	-0.0000	-0.0003	-0.0002	-0.0001	0.0004	0.0002	0.0000
Beta	-0.1581	0.7567	1.3973	0.9680	0.5678	1.3097	1.1885	1.0232
Alpha Robust	0.0002	0.0004	-0.0007	-0.0002	0.0004	0.0003	-0.0002	0.0004
Beta Robust	-0.1832	0.7532	1.3621	0.9769	0.5676	1.2352	1.1101	1.0433
Beta+	-0.1174	0.7014	1.5163	0.9332	0.6415	1.3386	1.2024	1.1207
Beta-	-0.2448	0.7075	1.1154	1.0232	0.5607	1.1065	1.2557	1.0538
Beta+ Robust	-0.1416	0.7315	1.4576	1.0280	0.6268	1.3432	1.1375	1.0888
Beta- Robust	-0.3032	0.6209	0.9889	0.9803	0.5605	0.9791	1.3130	1.0579
R-squared	0.1560	0.3672	0.1999	0.8159	0.4215	0.5204	0.7389	0.8097
R-squared Robust	0.1384	0.3309	0.1921	0.7294	0.4079	0.4903	0.6804	0.7590
Annualized Alpha	0.0290	-0.0101	-0.0814	-0.0575	-0.0252	0.1115	0.0622	0.0123
Correlation	-0.3950	0.6059	0.4471	0.9033	0.6492	0.7214	0.8596	0.8998
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1643	0.1383	0.3818	0.0623	0.1073	0.1751	0.0989	0.0672
Active Premium	-0.0970	-0.0456	-0.1257	-0.0687	-0.0759	0.1399	0.0836	0.0135
Information Ratio	-0.5902	-0.3299	-0.3293	-1.1023	-0.7079	0.7989	0.8456	0.2009
Treynor Ratio	-0.0613	0.0807	-0.0136	0.0392	0.0541	0.1883	0.1601	0.1174

Table 26: Complete CAPM statistics for 2017.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0003	-0.0002	-0.0002	0.0001	0.0001	-0.0007	-0.0002	0.0003
Beta	-0.2221	0.5271	0.3836	0.9142	0.4381	0.8311	1.3543	1.1629
Alpha Robust	0.0003	0.0001	0.0012	0.0001	0.0002	-0.0007	-0.0005	0.0007
Beta Robust	-0.2580	0.5231	0.3768	0.9186	0.4899	0.8203	1.4330	1.0811
Beta+	-0.2293	0.4960	0.2935	0.8684	0.5265	0.8846	1.4900	1.1036
Beta-	-0.1554	0.4383	0.0452	0.8805	0.3478	0.6378	1.3107	1.2966
Beta+ Robust	-0.2866	0.5122	0.0734	0.8230	0.5815	0.8688	1.5238	1.0738
Beta- Robust	-0.1720	0.4186	0.1629	0.8645	0.3639	0.5810	1.4088	1.3715
R-squared	0.1346	0.1307	0.0118	0.6138	0.1605	0.1915	0.5250	0.6090
R-squared Robust	0.1555	0.1074	0.0130	0.5821	0.2003	0.2042	0.4551	0.5665
Annualized Alpha	0.0726	-0.0382	-0.0468	0.0249	0.0350	-0.1553	-0.0530	0.0696
Correlation	-0.3669	0.3615	0.1084	0.7835	0.4007	0.4377	0.7246	0.7804
Correlation p-value	0.0000	0.0000	0.0859	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.0940	0.1006	0.2495	0.0510	0.0802	0.1199	0.0933	0.0661
Active Premium	-0.1769	-0.1477	-0.2108	0.0092	-0.0837	-0.2251	0.0076	0.1200
Information Ratio	-1.8822	-1.4690	-0.8450	0.1795	-1.0435	-1.8781	0.0814	1.8163
Treynor Ratio	-0.1246	0.1079	-0.0162	0.2338	0.2759	-0.0247	0.1567	0.2792

Table 27: Complete CAPM statistics for 2018.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0000	-0.0001	-0.0007	0.0003	-0.0002	-0.0006	-0.0004	0.0002
Beta	-0.0945	0.5377	0.6524	1.0783	0.5633	0.9845	1.0038	1.3130
Alpha Robust	0.0000	-0.0002	0.0006	0.0004	-0.0002	-0.0004	-0.0006	0.0002
Beta Robust	-0.0928	0.5302	0.5930	1.0862	0.5784	0.9702	1.0119	1.3039
Beta+	-0.0664	0.5118	0.3093	1.1514	0.5425	0.9538	1.0403	1.3794
Beta-	-0.1100	0.5967	0.4954	1.0553	0.6389	0.8849	1.0780	1.2448
Beta+ Robust	-0.0747	0.5138	-0.2492	1.1628	0.5440	0.9393	1.0660	1.3952
Beta- Robust	-0.1087	0.5671	0.6184	1.0776	0.7035	0.8801	1.0717	1.2376
R-squared	0.1538	0.3151	0.1330	0.8747	0.4241	0.5495	0.7371	0.8892
R-squared Robust	0.1444	0.2370	0.0986	0.7822	0.3567	0.4645	0.6467	0.7965
Annualized Alpha	0.0061	-0.0305	-0.1623	0.0743	-0.0508	-0.1372	-0.0856	0.0545
Correlation	-0.3922	0.5614	0.3647	0.9353	0.6513	0.7413	0.8585	0.9430
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1871	0.1537	0.2851	0.0696	0.1321	0.1494	0.1005	0.0937
Active Premium	0.0795	-0.0031	-0.1599	0.0601	-0.0217	-0.1357	-0.0843	0.0207
Information Ratio	0.4249	-0.0202	-0.5609	0.8640	-0.1644	-0.9083	-0.8394	0.2213
Treynor Ratio	-0.1132	-0.1337	-0.3505	-0.0080	-0.1607	-0.2077	-0.1525	-0.0366

Table 28: Complete CAPM statistics for 2019.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0005	0.0005	0.0002	-0.0001	0.0004	-0.0007	-0.0000	0.0002
Beta	-0.2096	0.4256	0.8587	1.0143	0.5314	1.0539	1.0514	1.3189
Alpha Robust	0.0006	0.0009	0.0014	0.0000	0.0007	-0.0010	-0.0000	0.0004
Beta Robust	-0.2031	0.4410	0.7564	1.0017	0.5082	1.0611	1.0519	1.2845
Beta+	-0.1549	0.5141	0.9871	1.0750	0.5184	0.9955	0.9678	1.3232
Beta-	-0.2730	0.3846	1.0035	0.9959	0.5959	1.0903	1.1577	1.3326
Beta+ Robust	-0.1609	0.4582	0.8369	1.0114	0.5501	1.0320	1.0236	1.3075
Beta- Robust	-0.2525	0.5092	0.7561	1.0073	0.4672	1.0459	1.1663	1.2421
R-squared	0.2447	0.1977	0.1235	0.8298	0.3927	0.4931	0.7334	0.8505
R-squared Robust	0.2390	0.1827	0.1206	0.7950	0.2995	0.4805	0.6288	0.7862
Annualized Alpha	0.1485	0.1478	0.0594	-0.0200	0.1040	-0.1606	-0.0047	0.0496
Correlation	-0.4947	0.4446	0.3515	0.9109	0.6267	0.7022	0.8564	0.9222
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1597	0.1304	0.2895	0.0581	0.1023	0.1351	0.0803	0.0806
Active Premium	-0.2246	-0.0257	-0.0265	-0.0235	-0.0362	-0.2043	0.0070	0.1783
Information Ratio	-1.4061	-0.1969	-0.0915	-0.4055	-0.3533	-1.5116	0.0875	2.2116
Treynor Ratio	-0.3934	0.6610	0.3267	0.2795	0.5097	0.0975	0.2987	0.3680

Table 29: Complete CAPM statistics for 2020.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0004	-0.0010	-0.0052	0.0003	-0.0002	-0.0026	-0.0010	0.0006
Beta	-0.1021	1.1119	0.9300	0.9485	0.7040	1.3961	1.2058	1.1069
Alpha Robust	0.0004	-0.0011	-0.0006	0.0001	-0.0003	-0.0039	-0.0014	0.0004
Beta Robust	-0.0861	1.0074	0.9761	0.9794	0.5644	1.2538	1.2092	1.1354
Beta+	-0.1079	1.0457	0.0775	0.8869	0.8074	1.1657	1.2379	1.1472
Beta-	-0.0932	1.2077	1.5371	0.9790	0.7063	1.4116	1.1950	1.0788
Beta+ Robust	-0.0761	1.0029	0.5693	0.9599	0.5409	0.9669	1.2572	1.1582
Beta- Robust	-0.0955	0.9973	1.0340	0.9789	0.5637	1.1689	1.1994	1.1040
R-squared	0.2439	0.8049	0.1908	0.9229	0.7618	0.6221	0.8373	0.8967
R-squared Robust	0.1725	0.5795	0.2023	0.8300	0.5472	0.4716	0.6812	0.7377
Annualized Alpha	0.1170	-0.2243	-0.7307	0.0799	-0.0376	-0.4797	-0.2174	0.1615
Correlation	-0.4939	0.8972	0.4368	0.9607	0.8728	0.7887	0.9151	0.9470
Correlation p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.3849	0.1926	0.6607	0.0962	0.1698	0.3992	0.1965	0.1347
Active Premium	-0.0484	-0.2670	-0.9007	0.0804	-0.0810	-0.5788	-0.2520	0.1824
Information Ratio	-0.1257	-1.3862	-1.3632	0.8361	-0.4773	-1.4498	-1.2824	1.3547
Treynor Ratio	-0.9059	-0.1134	-0.8169	0.2334	0.0851	-0.3136	-0.0921	0.2921

Table 30: Complete CAPM statistics for 2021.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0001	0.0007	0.0011	-0.0001	0.0003	0.0007	0.0004	0.0000
Beta	-0.0338	0.7227	0.9068	1.1299	0.4134	1.0523	0.9408	1.2381
Alpha Robust	-0.0001	0.0009	0.0023	-0.0001	0.0004	0.0003	0.0003	0.0001
Beta Robust	-0.0407	0.7006	0.8317	1.1001	0.3905	1.1084	0.9696	1.2586
Beta+	-0.0568	0.6749	0.6879	0.9975	0.3386	1.0885	1.0449	1.3101
Beta-	0.0393	0.7908	0.9266	1.2506	0.5150	0.8164	0.9214	1.2105
Beta+ Robust	-0.0503	0.6663	1.0935	0.9432	0.3028	1.2600	1.0864	1.3231
Beta- Robust	0.0237	0.7745	0.1556	1.2573	0.4908	0.8244	1.0206	1.2633
R-squared	0.0075	0.4452	0.1409	0.7221	0.2570	0.2358	0.4611	0.7659
R-squared Robust	0.0102	0.3987	0.1147	0.6590	0.1894	0.2528	0.4613	0.7478
Annualized Alpha	-0.0249	0.1978	0.3353	-0.0163	0.0715	0.1888	0.0957	0.0111
Correlation	-0.0868	0.6672	0.3754	0.8497	0.5069	0.4856	0.6791	0.8751
Correlation p-value	0.1687	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1502	0.1161	0.3049	0.0970	0.1246	0.2578	0.1386	0.0986
Active Premium	-0.2835	0.1516	0.3106	0.0083	-0.0775	0.2036	0.0895	0.0733
Information Ratio	-1.8868	1.3059	1.0186	0.0859	-0.6218	0.7894	0.6457	0.7437
Treynor Ratio	1.0026	0.5551	0.6178	0.2283	0.4164	0.4306	0.3604	0.2608

Table 31: Complete CAPM statistics for 2022.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0006	-0.0005	0.0011	-0.0007	0.0004	0.0025	0.0003	-0.0002
Beta	0.0795	0.8167	0.1888	1.2745	0.4998	0.6282	0.8706	1.2697
Alpha Robust	-0.0008	-0.0007	0.0026	-0.0006	0.0007	0.0036	0.0002	-0.0001
Beta Robust	0.0516	0.7828	0.2632	1.2895	0.4768	0.5983	0.8711	1.2721
Beta+	0.1626	0.9625	0.4641	1.1232	0.5049	0.5206	0.9738	1.3316
Beta-	0.0187	0.8280	0.3078	1.3336	0.5833	0.9980	0.8532	1.2044
Beta+ Robust	0.0650	0.8138	0.5579	1.1870	0.4983	0.4762	0.9763	1.3560
Beta- Robust	0.0133	0.8205	0.2577	1.3384	0.5069	0.8860	0.8200	1.2011
R-squared	0.0378	0.6776	0.0120	0.8537	0.4970	0.1955	0.7861	0.9300
R-squared Robust	0.0160	0.6371	0.0236	0.8415	0.4940	0.1761	0.7827	0.9177
Annualized Alpha	-0.1376	-0.1146	0.3262	-0.1583	0.1145	0.8886	0.0810	-0.0488
Correlation	0.1944	0.8231	0.1098	0.9239	0.7050	0.4422	0.8866	0.9643
Correlation p-value	0.0019	0.0000	0.0820	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.2497	0.1473	0.4705	0.1479	0.1763	0.3300	0.1174	0.1096
Active Premium	0.0652	-0.0605	0.3824	-0.1788	0.2051	0.7676	0.0884	-0.0962
Information Ratio	0.2611	-0.4106	0.8127	-1.2088	1.1630	2.3261	0.7530	-0.8784
Treynor Ratio	-1.9730	-0.3460	0.8487	-0.3145	-0.0341	0.8683	-0.1536	-0.2507

Table 32: Complete CAPM statistics for 2023.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0001	-0.0006	-0.0005	0.0002	-0.0005	-0.0006	-0.0005	0.0007
Beta	0.0295	1.0937	0.4015	1.2768	0.4610	0.7042	0.9906	1.1651
Alpha Robust	-0.0001	-0.0011	-0.0001	0.0001	-0.0003	-0.0002	-0.0001	0.0005
Beta Robust	0.0521	1.0581	0.4313	1.2659	0.4279	0.6725	0.9314	1.1633
Beta+	0.1644	1.1835	0.3371	1.3205	0.5074	0.5275	0.9452	1.2319
Beta-	-0.0872	1.2110	0.1032	1.3143	0.3748	0.5559	1.0775	1.0530
Beta+ Robust	0.1576	1.0257	0.4098	1.3153	0.4755	0.6927	0.9563	1.2429
Beta- Robust	-0.0059	1.2011	0.0730	1.2892	0.3329	0.4960	0.9488	1.0406
R-squared	0.0018	0.5378	0.0298	0.7462	0.2991	0.1724	0.6789	0.7246
R-squared Robust	0.0048	0.4983	0.0329	0.7397	0.2728	0.1720	0.6592	0.7446
Annualized Alpha	0.0253	-0.1366	-0.1133	0.0398	-0.1125	-0.1496	-0.1099	0.1940
Correlation	0.0424	0.7333	0.1727	0.8638	0.5469	0.4152	0.8240	0.8512
Correlation p-value	0.5037	0.0000	0.0061	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1610	0.1373	0.3193	0.1071	0.1198	0.2119	0.0919	0.0994
Active Premium	-0.2193	-0.1588	-0.3212	0.1199	-0.2665	-0.2729	-0.1438	0.2871
Information Ratio	-1.3617	-1.1565	-1.0061	1.1187	-2.2254	-1.2877	-1.5650	2.8877
Treynor Ratio	0.9398	0.0806	-0.1849	0.2873	-0.0424	-0.0368	0.1041	0.4583

Table 33: Complete CAPM statistics for 2024.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0001	-0.0004	0.0004	-0.0001	0.0003	-0.0002	0.0004	-0.0005
Beta	0.0473	0.6374	0.0785	1.2233	0.2402	0.4605	0.7266	1.5129
Alpha Robust	0.0000	-0.0003	0.0009	-0.0005	0.0001	0.0002	0.0003	-0.0002
Beta Robust	0.0931	0.6787	0.0625	1.2230	0.3090	0.4614	0.6922	1.4421
Beta+	-0.0425	0.2665	0.2770	1.0700	0.1643	0.5306	0.9059	1.5018
Beta-	-0.0756	0.6348	0.5959	1.3197	0.2471	0.7247	0.8137	1.4527
Beta+ Robust	0.0827	0.4871	0.2354	1.1789	0.3093	0.2677	0.6303	1.4692
Beta- Robust	-0.0428	0.4210	0.5699	1.3642	0.3190	0.7290	0.8619	1.4666
R-squared	0.0083	0.2595	0.0014	0.7157	0.1000	0.1106	0.4434	0.7728
R-squared Robust	0.0247	0.2567	0.0007	0.6932	0.1174	0.0767	0.3940	0.7015
Annualized Alpha	-0.0185	-0.0936	0.1058	-0.0272	0.0687	-0.0440	0.1168	-0.1167
Correlation	0.0910	0.5094	0.0378	0.8460	0.3162	0.3325	0.6659	0.8791
Correlation p-value	0.1488	0.0000	0.5494	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.1399	0.1466	0.2929	0.1035	0.1351	0.1823	0.1108	0.1248
Active Premium	-0.2331	-0.2000	-0.1376	0.0123	-0.1037	-0.1861	0.0650	-0.0399
Information Ratio	-1.6663	-1.3643	-0.4697	0.1189	-0.7679	-1.0210	0.5871	-0.3199
Treynor Ratio	-0.2325	0.0347	1.0765	0.1916	0.4926	0.0781	0.3951	0.1204

Table 34: Complete CAPM statistics for 2025.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	0.0003	-0.0002	-0.0006	-0.0004	-0.0001	-0.0001	0.0000	0.0000
Beta	-0.0243	0.5715	0.4763	1.1240	0.2405	0.8131	0.8413	1.3308
Alpha Robust	0.0003	0.0001	-0.0009	-0.0008	0.0001	0.0008	0.0003	0.0001
Beta Robust	-0.0234	0.5739	0.5578	1.1800	0.1230	0.5923	0.7920	1.3127
Beta+	-0.0343	0.5854	0.6091	1.1461	0.3277	0.7983	0.7818	1.2645
Beta-	-0.0733	0.5849	1.0135	0.9927	0.1893	1.2628	0.9059	1.3606
Beta+ Robust	-0.0422	0.5780	0.6392	1.1448	0.1793	0.8020	0.7797	1.2660
Beta- Robust	-0.0652	0.6281	1.0801	1.1557	-0.0107	1.3551	0.8016	1.4335
R-squared	0.0070	0.4185	0.0890	0.8099	0.1124	0.3979	0.7210	0.8763
R-squared Robust	0.0047	0.2629	0.0569	0.7141	0.0229	0.1384	0.5207	0.7522
Annualized Alpha	0.0827	-0.0465	-0.1417	-0.1046	-0.0189	-0.0361	0.0100	0.0062
Correlation	-0.0834	0.6469	0.2983	0.9000	0.3352	0.6308	0.8491	0.9361
Correlation p-value	0.1876	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Tracking Error	0.2050	0.1538	0.3104	0.1075	0.1958	0.1959	0.1053	0.1155
Active Premium	-0.0694	-0.1195	-0.2650	-0.1106	-0.1376	-0.0866	-0.0165	0.0452
Information Ratio	-0.3383	-0.7770	-0.8538	-1.0287	-0.7029	-0.4418	-0.1565	0.3913
Treynor Ratio	-3.1715	0.0470	-0.2492	0.0318	0.0362	0.0735	0.1543	0.1439

Table 35: Complete CAPM statistics for 2026, based on the partial final endpoint interval.

Statistic	IEF	VNQ	USO	XLY	XLP	XLE	XLF	XLK
Alpha	-0.0000	0.0006	0.0095	-0.0010	0.0007	0.0045	-0.0010	-0.0000
Beta	0.0849	0.5150	-1.5587	1.1513	0.0925	-0.0751	0.8529	1.5393
Alpha Robust	-0.0000	0.0010	0.0080	-0.0007	0.0008	0.0052	-0.0007	0.0007
Beta Robust	0.0864	0.4477	-1.1127	1.2404	0.0746	-0.0012	0.8045	1.5171
Beta+	0.1136	0.3915	-2.1532	1.1053	0.2098	-0.2445	0.7334	1.5575
Beta-	0.1349	0.8109	-1.4185	1.3071	0.0644	0.5364	0.7859	1.3508
Beta+ Robust	0.1083	0.3600	-1.2119	2.4877	0.3544	-0.1974	0.7156	1.4507
Beta- Robust	0.1363	0.7616	-0.4892	1.3079	0.0480	0.5378	0.7668	1.3982
R-squared	0.0569	0.2753	0.1645	0.6985	0.0077	0.0024	0.4694	0.8256
R-squared Robust	0.0437	0.2377	0.0911	0.6970	0.0055	0.0000	0.4439	0.7934
Annualized Alpha	-0.0016	0.1760	9.9429	-0.2319	0.2001	2.0778	-0.2248	-0.0094
Correlation	0.2385	0.5247	-0.4056	0.8358	0.0877	-0.0487	0.6851	0.9087
Correlation p-value	0.0538	0.0000	0.0007	0.0000	0.4840	0.6976	0.0000	0.0000
Tracking Error	0.1400	0.1383	0.6219	0.1104	0.1988	0.2687	0.1315	0.1273
Active Premium	0.1176	0.2203	10.7174	-0.2206	0.3039	2.1664	-0.1861	-0.0818
Information Ratio	0.8400	1.5930	17.2326	-1.9979	1.5284	8.0622	-1.4153	-0.6425
Treynor Ratio	-0.1636	0.1724	-6.7917	-0.3058	1.8627	-27.0952	-0.3724	-0.1386

C Complete submitted R code

This is the complete file submitted for the homework, including the original question comments, diagnostic expressions, and filled-in solutions.

```
#################################
### FRE7241 Homework #1 due at 6PM Tuesday November 3
#################################
# Max score 80pts

# Please write in this file the R code needed to perform
# the tasks below, rename it to your_name_hw1.R,
# and upload the file to Brightspace

# Student: Panagiotis Housos
# Completed solution. Plots are saved in hw1_results/ in the working directory.
# Methods follow strategies_univariate.pdf (slides 14-19), functions.pdf,
# dates_time_series.pdf, and the CAPM examples in data_management.pdf.

############## Part I
# Summary: Calculate the optimal weights for a portfolio
# of stocks and bonds, to maximize the risk-adjusted
# wealth, for different holding periods.

# Disclaimer:
# The comments and conclusions below are based on
# 20 years of very positive stock and bond returns,
# when stocks and bonds have both been in a secular
# bull market.
# The conclusions would not hold if stocks and bonds
# had suffered from a bear market (losses).

## Run all the setup code below

library(rutils)
# Coerce the log prices from xts time series to matrix
pricev <- na.omit(rutils::etfenv$prices[, c("VTI", "IEF")])
pricev <- log(zoo::coredata(pricev))
nrows <- NROW(pricev)


## End of setup code


# 1. (10pts)
# Sample the start dates for the bootstrap.
# Run the code below.

# Follow the lecture convention: holdp counts price observations in a block,
# so the first and last prices are separated by holdp - 1 return intervals.
holdp <- 10 * 252
nboot <- 1000
set.seed(1121, "Mersenne-Twister", sample.kind = "Rejection")
# Include every valid block start, including the final possible block.
startd <- sample.int(nrows - holdp + 1L, size = nboot, replace = TRUE)


# Calculate a matrix of 1000 simulated VTI and IEF returns,
# for the 10 year holding period.

# Sample the same historical block for both ETFs to preserve dependence.
retm <- t(pricev[startd + holdp - 1L, , drop = FALSE] -
          pricev[startd, , drop = FALSE])


# You should get the outputs:
dim(retm)
# [1] 2 1000
retm[, 1:5]
```

```
#         [,1]      [,2]      [,3]      [,4]     [,5]
# VTI 1.3889156 1.1463289 0.7314036 1.5094435 1.11841720
# IEF 0.4327528 0.1021709 0.5198690 0.2757054 0.08607574


# Specify the weight for VTI.
# Run the code below.

weightv <- 0.5

# Calculate the vector of wealths for retm and weightv.
# You can use the functions drop(), c(), and the %*% operator.

# "Wealth" here is the weighted cumulative log return used in the lectures.
wealthv <- drop(c(weightv, 1 - weightv) %*% retm)


# You should get the outputs:
is.vector(wealthv)
# TRUE
NROW(wealthv)
# [1] 1000
wealthv[1:5]
# [1] 0.9108342 0.6242499 0.6256363 0.8925745 0.6022465


# 2. (10pts)
# Define a risk-adjusted wealth function, which accepts
# two arguments: retm and weightv, and returns the
# negative of the risk-adjusted wealth value.

riskretfun <- function(retm, weightv) {
  wealthv <- drop(c(weightv, 1 - weightv) %*% retm)
  -mean(wealthv) / sd(wealthv)
}


# You should get the output:
riskretfun(retm, weightv)
# [1] -6.934682

# Calculate the optimal VTI weight using the function
# optimize() and the function riskretfun().

# Minimize the negative mean/SD ratio over fully invested long-only weights.
weightv <- optimize(riskretfun, interval = c(0, 1), retm = retm)$minimum


# You should get the output:
weightv
# [1] 0.4543241


# 3. (10pts)
# Define a vector of holding periods.
# Run the code below.

holdv <- round(nrows * seq(0.1, 0.5, 0.1))


# Calculate a named list of 1000 simulated VTI and
# IEF returns, one for each holding period in holdv.

set.seed(1121, "Mersenne-Twister", sample.kind="Rejection")

retl <- lapply(holdv, function(holdp) {
  startd <- sample.int(nrows - holdp + 1L, size = nboot, replace = TRUE)
  t(pricev[startd + holdp - 1L, , drop = FALSE] -
    pricev[startd, , drop = FALSE])
})
```

```
names(retl) <- paste0("hold", holdv)


# You should get the outputs:
NROW(retl)
# [1] 5
class(retl)
# [1] "list"
names(retl)
# [1] "hold596"  "hold1192" "hold1789" "hold2385" "hold2981"
dim(retl[[1]])
# [1] 2 1000
# You should get output similar to:
retl[[1]][, 1:5]
#         [,1]       [,2]       [,3]       [,4]      [,5]
# VTI 0.26369527 0.3695579 0.1482638  0.2195997 0.2832994
# IEF 0.09298308 0.2352719 0.1463518 -0.2017072 0.2119880

# Your values may be slightly different, because
# they're from a simulation.

# Calculate a named vector of optimal VTI weights,
# for each holding period in holdv.
# Hint: Perform an sapply() loop over the list retl.
# You must use the function optimize().

weightv <- sapply(retl, function(retm) {
  optimize(riskretfun, interval = c(0, 1), retm = retm)$minimum
})


# You should get output similar to:
weightv
#   hold596  hold1192  hold1789  hold2385  hold2981
# 0.4127018 0.3597272 0.4160783 0.4588359 0.4322657

# Your values may be slightly different, because
# they're from a simulation.

# Note that the optimal VTI weights increase with
# longer holding periods.
# Except for the shortest holding period, which is
# large.
# This may be because of higher correlations between
# stocks and bonds for very short holding periods.


# 4. (10pts)
# Calculate a list of portfolio wealths, for each
# holding period in holdv.
# Hint: Perform an lapply() loop over names(retl).

wealthl <- lapply(names(retl), function(name) {
  weight <- weightv[[name]]
  drop(c(weight, 1 - weight) %*% retl[[name]])
})
names(wealthl) <- names(retl)


# You should get the outputs:
NROW(wealthl)
# [1] 5
class(wealthl)
# [1] "list"
is.vector(wealthl[[1]])
# [1] TRUE
# You should get output similar to:
wealthl[[1]][1:5]
# [1]  0.16343630  0.29069195  0.14714087 -0.02783308  0.24141838

```

```
205
206 # Plot the distributions of the portfolio wealths,
207 # for the shortest and longest holding periods.
208
209 # Your plot should be similar to portf_wealth_longshort.png
210
211 denslong <- density(wealthl[[which.max(holdv)]])
212 densshort <- density(wealthl[[which.min(holdv)]])
213 dir.create("hw1_results", showWarnings = FALSE)
214 png(file.path("hw1_results", "portf_wealth_longshort.png"),
215     width = 1500, height = 1250, res = 160)
216 plot(denslong, col = "blue", lwd = 3,
217      xlim = range(denslong$x, densshort$x),
218      ylim = c(0, max(denslong$y, densshort$y)),
219      main = "Wealth Distributions for Long and Short Holding Periods",
220      cex.main = 1.1, xlab = "wealth", ylab = "Density")
221 lines(densshort, col = "green", lwd = 3)
222 legend("topright", legend = c("Long", "Short"),
223        col = c("blue", "green"), lwd = 6, bty = "n", inset = 0.05)
224 invisible(dev.off())
225
226
227 ############## Part II
228 # Summary: Calculate a matrix of the best performing ETFs
229 # in each year.  Create a scatterplot of alphas for the
230 # years 2008 and 2009.
231
232 ## Run the setup code below
233
234 library(rutils)
235 library(PerformanceAnalytics)
236 symbolv <- c("VTI", "IEF", "VNQ", "USO", "XLY", "XLP", "XLE", "XLF", "XLK")
237 retp <- rutils::etfenv$returns[, symbolv]
238 retp <- na.omit(zoo::na.locf(retp))
239
240 ## End of setup code.
241
242
243 # 1. (10pts)
244 # Create a vector of yearly end points of the returns.
245 # You can use the function rutils::calc_endpoints().
246
247 endd <- rutils::calc_endpoints(retp, interval = "years")
248
249
250 # You should get the output:
251 endd
252 #  [1]    0  183  434  687  939 1191 1443 1693 1945 2197 2449 2701 2952 3203 3455 3708
253 # [17] 3960 4211 4461 4713 4963 5028
254
255 # Select the returns for the symbolv, and for the dates between
256 # the last two end points.
257 # You can use the function NROW().
258
259 # Include both endpoint rows, as illustrated by the assignment's retsub.
260 retsub <- retp[seq.int(endd[NROW(endd) - 1L], endd[NROW(endd)]), symbolv]
261
262
263 # You should get the outputs:
264 head(retsub)
265 #                    VTI           IEF           VNQ          USO          XLY
266 # 2025-12-31 -0.0076658326 -0.0033222622 -0.0082156784 -0.008351380 -0.007924302
267 # 2026-01-02  0.0030971763 -0.0008322930  0.0003389639 -0.002896034 -0.008916614
268 # 2026-01-05  0.0074946659  0.0030137721  0.0022568279  0.018106544  0.016342287
269 # 2026-01-06  0.0069701039 -0.0007266311  0.0072997124 -0.024653451  0.006380268
270 # 2026-01-07 -0.0034054579  0.0018674142 -0.0097825034 -0.010565029 -0.001405133
271 # 2026-01-08  0.0002646319 -0.0030103308  0.0088869475  0.039765233  0.016569980
272 #                    XLP          XLE          XLF           XLK
273 # 2025-12-31 -0.006032233 -0.005576015 -0.007457970 -0.0099523940
274 # 2026-01-02  0.000128725  0.020806417  0.002917049  0.0022895212
```

```
# 2026-01-05 -0.004385972  0.026800826  0.021610782  0.0022151470
# 2026-01-06  0.001550188 -0.027019908  0.004798730  0.0139391827
# 2026-01-07 -0.011032625 -0.011237311 -0.014106117 -0.0008186098
# 2026-01-08  0.023475862  0.030979790  0.005201339 -0.0157516068
tail(retsub)
#                    VTI           IEF         VNQ          USO           XLY
# 2026-03-30 -0.0045136687  0.0070574897 0.003785928  0.044332733 -0.0001892685
# 2026-03-31  0.0288720271  0.0017828121 0.015565843 -0.020072247  0.0309379874
# 2026-04-01  0.0075460338 -0.0008682939 0.003601174 -0.025146547  0.0074961499
# 2026-04-02  0.0016074192  0.0023121398 0.013500902  0.105666698 -0.0151413767
# 2026-04-06  0.0044686269 -0.0026278462 0.001439726  0.007368378  0.0081956350
# 2026-04-07  0.0006762572  0.0025228648 0.001437656 -0.006208958 -0.0117154608
#                   XLP          XLE         XLF          XLK
# 2026-03-30  0.001222046 -0.009637081 0.011438203 -0.018802512
# 2026-03-31  0.001220554 -0.011361914 0.020669927  0.041480601
# 2026-04-01 -0.006363213 -0.038098261 0.001416861  0.015010924
# 2026-04-02  0.005264781  0.004736940 0.001818733  0.007973464
# 2026-04-06  0.009358926  0.007231176 0.007041574  0.005792442
# 2026-04-07 -0.017081919  0.008010724 0.000000000  0.004740901

# Perform an lapply() loop over the neighboring end points
# of endd.
# Inside the loop, select the returns for the symbolv and
# the neighboring end points.
# Then for the selected returns, calculate a data frame of
# statistics using table.CAPM(), and using VTI as the
# benchmark asset "Rb".
# Simplify the column names and return the data frame.
# The output should be a list of data frames.
# You can use the functions lapply(), sapply(),
# PerformanceAnalytics::table.CAPM(), colnames(),
# strsplit(), NROW(), and an anonymous function.
# You don't need to use all of these functions.
# You can use any functions you choose.
# If you're getting an error when running table.CAPM(),
# then install the package RobStatTM:
#   install.packages("RobStatTM")

capml <- lapply(seq_len(NROW(endd) - 1L), function(i) {
  # The assignment's endpoint intervals share the boundary observation.
  # max(1L, ...) replaces the initial zero endpoint with the first row.
  rows <- seq.int(max(1L, endd[i]), endd[i + 1L])
  retsub <- retp[rows, symbolv]
  capm <- PerformanceAnalytics::table.CAPM(
    Ra = retsub[, symbolv[-1]], Rb = retsub[, "VTI"], Rf = 0, scale = 252
  )
  colnames(capm) <- sub(" to .*$", "", colnames(capm))
  capm
})

# You should get the outputs:
is.list(capml)
# [1] TRUE
capml[[1]]
#                         IEF    VNQ     USO    XLY    XLP     XLE    XLF     XLK
# Alpha                0.0003 0.0007 -0.0016 0.0002 0.0004 -0.0004 0.0003 -0.0002
# Beta                 0.0326 0.9094  0.0865 0.9956 0.5466  1.2493 0.9259  1.1092
# Alpha Robust         0.0003 0.0006 -0.0015 0.0003 0.0005 -0.0001 0.0002 -0.0002
# Beta Robust          0.0523 0.9582  0.0881 0.9822 0.5670  1.2525 0.8963  1.1390
# Beta+                0.0831 0.8206  0.6261 1.0077 0.3818  1.4792 0.8758  1.0898
# Beta-                0.0188 0.6517  0.0731 0.9016 0.6239  1.5042 0.9472  0.9853
# Beta+ Robust         0.0868 0.8887  0.6261 0.9673 0.3920  1.4617 0.8617  1.0265
# Beta- Robust         0.0639 0.7227  0.0835 0.8966 0.6247  1.5130 0.9279  1.0427
# R-squared            0.0069 0.4583  0.0015 0.7497 0.4770  0.3022 0.7175  0.7213
# R-squared Robust     0.0183 0.4976  0.0015 0.7283 0.4658  0.3144 0.6502  0.7006
# Annualized Alpha     0.0689 0.1926 -0.3242 0.0632 0.1090 -0.0886 0.0769 -0.0604
# Correlation          0.0832 0.6770  0.0383 0.8658 0.6906  0.5498 0.8471  0.8493
# Correlation p-value  0.2631 0.0000  0.6064 0.0000 0.0000  0.0000 0.0000  0.0000
# Tracking Error       0.1144 0.1089  0.2668 0.0631 0.0801  0.2099 0.0642  0.0765
# Active Premium      -0.0577 0.1957 -0.4673 0.0684 0.0550 -0.0928 0.0739 -0.0578
```

```
# Information Ratio   -0.5040 1.7976 -1.7512 1.0842 0.6870 -0.4421 1.1511 -0.7553
# Treynor Ratio        2.2196 0.3582 -3.8978 0.1993 0.3385  0.0298 0.2203  0.0651

# Assign names to the list using the years corresponding to the endd.
# You can use the functions names(), format() with the "%Y" format,
# and zoo::index(),

names(capml) <- format(zoo::index(retp)[endd[-1L]], "%Y")


# You should get the outputs:
names(capml)
#  [1] "2006" "2007" "2008" "2009" "2010" "2011" "2012" "2013" "2014" "2015" "2016"
# [12] "2017" "2018" "2019" "2020" "2021" "2022" "2023" "2024" "2025" "2026"


# 2. (10pts)
# Perform an sapply() loop over "capml".
# Inside the loop extract the data frame row called
# "Annualized Alpha", coerce it to a vector using unlist(),
# and return the vector.
# You can use the functions sapply(), unlist(),
# and an anonymous function,

alphav <- sapply(capml, function(capm) {
  unlist(capm["Annualized Alpha", ])
})


# alphav should be a matrix of annual alphas like this:
dim(alphav)
# [1]  8 21
alphav
#       2006    2007    2008    2009    2010    2011    2012    2013    2014    2015    2016
# IEF  0.0689  0.1137  0.1205 -0.0488  0.1229  0.1639  0.0790 -0.0470  0.1114  0.0152  0.0290
# VNQ  0.1926 -0.2144  0.2848 -0.2019  0.0418  0.0730  0.0489 -0.2262  0.2165  0.0116 -0.0101
# USO -0.3242  0.4603 -0.4475 -0.0128 -0.1545 -0.0072 -0.2647 -0.1209 -0.4477 -0.4537 -0.0814
# XLY  0.0632 -0.1814  0.0540  0.0706  0.0763  0.0481  0.0654  0.0602 -0.0286  0.1013 -0.0575
# XLP  0.1090  0.0921  0.0858  0.0086  0.0341  0.1348  0.0166  0.0090  0.0800  0.0608 -0.0252
# XLE -0.0886  0.2754  0.0866 -0.0996  0.0129  0.0183 -0.1196 -0.0710 -0.1912 -0.2141  0.1115
# XLF  0.0769 -0.2474 -0.0868 -0.3040 -0.0766 -0.1798  0.0701 -0.0409  0.0257 -0.0217  0.0622
# XLK -0.0604  0.1025 -0.1060  0.2115 -0.0430  0.0164 -0.0115 -0.0142  0.0523  0.0502  0.0123
#       2017    2018    2019    2020    2021    2022    2023    2024    2025    2026
# IEF  0.0726  0.0061  0.1485  0.1170 -0.0249 -0.1376  0.0253 -0.0185  0.0827 -0.0016
# VNQ -0.0382 -0.0305  0.1478 -0.2243  0.1978 -0.1146 -0.1366 -0.0936 -0.0465  0.1760
# USO -0.0468 -0.1623  0.0594 -0.7307  0.3353  0.3262 -0.1133  0.1058 -0.1417  9.9429
# XLY  0.0249  0.0743 -0.0200  0.0799 -0.0163 -0.1583  0.0398 -0.0272 -0.1046 -0.2319
# XLP  0.0350 -0.0508  0.1040 -0.0376  0.0715  0.1145 -0.1125  0.0687 -0.0189  0.2001
# XLE -0.1553 -0.1372 -0.1606 -0.4797  0.1888  0.8886 -0.1496 -0.0440 -0.0361  2.0778
# XLF -0.0530 -0.0856 -0.0047 -0.2174  0.0957  0.0810 -0.1099  0.1168  0.0100 -0.2248
# XLK  0.0696  0.0545  0.0496  0.1615  0.0111 -0.0488  0.1940 -0.1167  0.0062 -0.0094

# 3. (10pts)
# Sort the last column of alphav in descending order,
# and extract the ETF names.
# You can use the functions names(), sort(), and NCOL().

names(sort(alphav[, NCOL(alphav)], decreasing = TRUE))


# You should get the output:
# [1] "USO" "XLE" "XLP" "VNQ" "IEF" "XLK" "XLF" "XLY"

# Sort all the columns of alphav in descending order,
# using an apply() loop over the columns of alphav, and
# extract the ETF names.
# You can use the functions apply(), sort(), names(),
# and an anonymous function,

namev <- apply(alphav, 2, function(alpha) {
```

```
  names(sort(alpha, decreasing = TRUE))
})


# You should get the outputs:
dim(namev)
# [1]  8 21
namev
#       2006  2007  2008  2009  2010  2011  2012  2013  2014  2015  2016  2017  2018  2019
# [1,] "VNQ" "USO" "VNQ" "XLK" "IEF" "IEF" "IEF" "XLY" "VNQ" "XLY" "XLE" "IEF" "XLY" "IEF"
# [2,] "XLP" "XLE" "IEF" "XLY" "XLY" "XLP" "XLF" "XLP" "IEF" "XLP" "XLF" "XLK" "XLK" "VNQ"
# [3,] "XLF" "IEF" "XLE" "XLP" "VNQ" "VNQ" "XLY" "XLK" "XLP" "XLK" "IEF" "XLP" "IEF" "XLP"
# [4,] "IEF" "XLK" "XLP" "USO" "XLP" "XLY" "VNQ" "XLF" "XLK" "IEF" "XLK" "XLY" "VNQ" "USO"
# [5,] "XLY" "XLP" "XLY" "IEF" "XLE" "XLE" "XLP" "IEF" "XLF" "VNQ" "VNQ" "VNQ" "XLP" "XLK"
# [6,] "XLK" "XLY" "XLF" "XLE" "XLK" "XLK" "XLK" "XLE" "XLY" "XLF" "XLP" "USO" "XLF" "XLF"
# [7,] "XLE" "VNQ" "XLK" "VNQ" "XLF" "USO" "XLE" "USO" "XLE" "XLE" "XLY" "XLF" "XLE" "XLY"
# [8,] "USO" "XLF" "USO" "XLF" "USO" "XLF" "USO" "VNQ" "USO" "USO" "USO" "XLE" "USO" "XLE"
#       2020  2021  2022  2023  2024  2025  2026
# [1,] "XLK" "USO" "XLE" "XLK" "XLF" "IEF" "USO"
# [2,] "IEF" "VNQ" "USO" "XLY" "USO" "XLF" "XLE"
# [3,] "XLY" "XLE" "XLP" "IEF" "XLP" "XLK" "XLP"
# [4,] "XLP" "XLF" "XLF" "XLF" "IEF" "XLP" "VNQ"
# [5,] "XLF" "XLP" "XLK" "XLP" "XLY" "XLE" "IEF"
# [6,] "VNQ" "XLK" "VNQ" "USO" "XLE" "VNQ" "XLK"
# [7,] "XLE" "XLY" "IEF" "VNQ" "VNQ" "XLY" "XLF"
# [8,] "USO" "IEF" "XLY" "XLE" "XLK" "USO" "XLY"


# 4. (10pts)
# Plot a scatterplot of the alphas for the years
# "2008" and "2009", and add labels with ETF names.
# You can use the functions plot(), rownames(),
# range(), and text().
# Hint: Use the parameters xlim and ylim in the
# function plot(), to enlarge the dimensions of
# the plot, and make room for the labels.
# Your plot should be similar to scatter_etfs.png

alpha2008 <- alphav[, "2008"]
alpha2009 <- alphav[, "2009"]
png(file.path("hw1_results", "scatter_etfs.png"),
    width = 1500, height = 1250, res = 160)
plot(alpha2008, alpha2009, pch = 1, cex = 1.2,
     xlim = extendrange(alpha2008, f = 0.20),
     ylim = extendrange(alpha2009, f = 0.15),
     main = "Scatterplot of ETF alphas in 2008 and 2009",
     xlab = "2008", ylab = "2009")
text(alpha2008, alpha2009, labels = rownames(alphav), pos = 4)
invisible(dev.off())


# Comment:
# The scatterplot of the ETF alphas in the years 2008 and 2009
# doesn't show any relationship between the alphas in those years.
# For some ETFs the positive or negative alphas persist, while for
# others, the alphas change radically from year to year.
```

D Complete assignment execution output

This record contains the outputs of the assignment's diagnostic expressions. Full additional scenario data and every annual CAPM table are also supplied in the accompanying CSV/RDS exports.

```
[1]    2 1000
        [,1]      [,2]      [,3]      [,4]       [,5]
VTI 1.3889156 1.1463289 0.7314036 1.5094435 1.11841720
IEF 0.4327528 0.1021709 0.5198690 0.2757054 0.08607574
[1] TRUE
[1] 1000
[1] 0.9108342 0.6242499 0.6256363 0.8925745 0.6022465
[1] -6.934682
[1] 0.4543241
[1] 5
[1] "list"
[1] "hold596"  "hold1192" "hold1789" "hold2385" "hold2981"
[1]    2 1000
         [,1]      [,2]      [,3]       [,4]      [,5]
VTI 0.26369527 0.3695579 0.1482638  0.2195997 0.2832994
IEF 0.09298308 0.2352719 0.1463518 -0.2017072 0.2119880
  hold596  hold1192  hold1789  hold2385  hold2981
0.4127018 0.3597272 0.4160783 0.4588359 0.4322657
[1] 5
[1] "list"
[1] TRUE
[1]  0.16343630  0.29069195  0.14714087 -0.02783308  0.24141838
 [1]    0  183  434  687  939 1191 1443 1693 1945 2197 2449 2701 2952 3203 3455
[16] 3708 3960 4211 4461 4713 4963 5028
                 VTI           IEF           VNQ           USO           XLY
2025-12-31 -0.0076658326 -0.0033222622 -0.0082156784 -0.008351380 -0.007924302
2026-01-02  0.0030971763 -0.0008322930  0.0003389639 -0.002896034 -0.008916614
2026-01-05  0.0074946659  0.0030137721  0.0022568279  0.018106544  0.016342287
2026-01-06  0.0069701039 -0.0007266311  0.0072997124 -0.024653451  0.006380268
2026-01-07 -0.0034054579  0.0018674142 -0.0097825034 -0.010565029 -0.001405133
2026-01-08  0.0002646319 -0.0030103308  0.0088869475  0.039765233  0.016569980
                 XLP           XLE           XLF           XLK
2025-12-31 -0.006032233 -0.005576015 -0.007457970 -0.0099523940
2026-01-02  0.000128725  0.020806417  0.002917049  0.0022895212
2026-01-05 -0.004385972  0.026800826  0.021610782  0.0022151470
2026-01-06  0.001550188 -0.027019908  0.004798730  0.0139391827
2026-01-07 -0.011032625 -0.011237311 -0.014106117 -0.0008186098
2026-01-08  0.023475862  0.030979790  0.005201339 -0.0157516068
                 VTI           IEF         VNQ           USO           XLY
2026-03-30 -0.0045136687  0.0070574897 0.003785928  0.044332733 -0.0001892685
2026-03-31  0.0288720271  0.0017828121 0.015565843 -0.020072247  0.0309379874
2026-04-01  0.0075460338 -0.0008682939 0.003601174 -0.025146547  0.0074961499
2026-04-02  0.0016074192  0.0023121398 0.013500902  0.105666698 -0.0151413767
2026-04-06  0.0044686269 -0.0026278462 0.001439726  0.007368378  0.0081956350
2026-04-07  0.0006762572  0.0025228648 0.001437656 -0.006208958 -0.0117154608
                 XLP           XLE         XLF           XLK
2026-03-30  0.001222046 -0.009637081 0.011438203 -0.018802512
2026-03-31  0.001220554 -0.011361914 0.020669927  0.041480601
2026-04-01 -0.006363213 -0.038098261 0.001416861  0.015010924
2026-04-02  0.005264781  0.004736940 0.001818733  0.007973464
2026-04-06  0.009358926  0.007231176 0.007041574  0.005792442
2026-04-07 -0.017081919  0.008010724 0.000000000  0.004740901
[1] TRUE
                       IEF    VNQ     USO    XLY    XLP     XLE    XLF     XLK
Alpha                0.0003 0.0007 -0.0016 0.0002 0.0004 -0.0004 0.0003 -0.0002
Beta                 0.0326 0.9094  0.0865 0.9956 0.5466  1.2493 0.9259  1.1092
Alpha Robust         0.0003 0.0006 -0.0015 0.0003 0.0005 -0.0001 0.0002 -0.0002
Beta Robust          0.0523 0.9582  0.0881 0.9822 0.5670  1.2525 0.8963  1.1390
Beta+                0.0831 0.8206  0.6261 1.0077 0.3818  1.4792 0.8758  1.0898
Beta-                0.0188 0.6517  0.0731 0.9016 0.6239  1.5042 0.9472  0.9853
Beta+ Robust         0.0868 0.8887  0.6261 0.9673 0.3920  1.4617 0.8617  1.0265
Beta- Robust         0.0639 0.7227  0.0835 0.8966 0.6247  1.5130 0.9279  1.0427
R-squared            0.0069 0.4583  0.0015 0.7497 0.4770  0.3022 0.7175  0.7213
R-squared Robust     0.0183 0.4976  0.0015 0.7283 0.4658  0.3144 0.6502  0.7006
Annualized Alpha     0.0689 0.1926 -0.3242 0.0632 0.1090 -0.0886 0.0769 -0.0604
Correlation          0.0832 0.6770  0.0383 0.8658 0.6906  0.5498 0.8471  0.8493
Correlation p-value  0.2631 0.0000  0.6064 0.0000 0.0000  0.0000 0.0000  0.0000
Tracking Error       0.1144 0.1089  0.2668 0.0631 0.0801  0.2099 0.0642  0.0765
Active Premium      -0.0577 0.1957 -0.4673 0.0684 0.0550 -0.0928 0.0739 -0.0578
Information Ratio   -0.5040 1.7976 -1.7512 1.0842 0.6870 -0.4421 1.1511 -0.7553
Treynor Ratio        2.2196 0.3582 -3.8978 0.1993 0.3385  0.0298 0.2203  0.0651
 [1] "2006" "2007" "2008" "2009" "2010" "2011" "2012" "2013" "2014" "2015"
[11] "2016" "2017" "2018" "2019" "2020" "2021" "2022" "2023" "2024" "2025"
[21] "2026"
[1]  8 21
       2006    2007    2008    2009    2010    2011    2012    2013    2014
IEF  0.0689  0.1137  0.1205 -0.0488  0.1229  0.1639  0.0790 -0.0470  0.1114
VNQ  0.1926 -0.2144  0.2848 -0.2019  0.0418  0.0730  0.0489 -0.2262  0.2165
USO -0.3242  0.4603 -0.4475 -0.0128 -0.1545 -0.0072 -0.2647 -0.1209 -0.4477
XLY  0.0632 -0.1814  0.0540  0.0706  0.0763  0.0481  0.0654  0.0602 -0.0286
XLP  0.1090  0.0921  0.0858  0.0086  0.0341  0.1348  0.0166  0.0090  0.0800
XLE -0.0886  0.2754  0.0866 -0.0996  0.0129  0.0183 -0.1196 -0.0710 -0.1912
XLF  0.0769 -0.2474 -0.0868 -0.3040 -0.0766 -0.1798  0.0701 -0.0409  0.0257
XLK -0.0604  0.1025 -0.1060  0.2115 -0.0430  0.0164 -0.0115 -0.0142  0.0523
       2015    2016    2017    2018    2019    2020    2021    2022    2023
IEF  0.0152  0.0290  0.0726  0.0061  0.1485  0.1170 -0.0249 -0.1376  0.0253
VNQ  0.0116 -0.0101 -0.0382 -0.0305  0.1478 -0.2243  0.1978 -0.1146 -0.1366
```

```
USO -0.4537 -0.0814 -0.0468 -0.1623  0.0594 -0.7307  0.3353  0.3262 -0.1133
XLY  0.1013 -0.0575  0.0249  0.0743 -0.0200  0.0799 -0.0163 -0.1583  0.0398
XLP  0.0608 -0.0252  0.0350 -0.0508  0.1040 -0.0376  0.0715  0.1145 -0.1125
XLE -0.2141  0.1115 -0.1553 -0.1372 -0.1606 -0.4797  0.1888  0.8886 -0.1496
XLF -0.0217  0.0622 -0.0530 -0.0856 -0.0047 -0.2174  0.0957  0.0810 -0.1099
XLK  0.0502  0.0123  0.0696  0.0545  0.0496  0.1615  0.0111 -0.0488  0.1940
       2024    2025    2026
IEF -0.0185  0.0827 -0.0016
VNQ -0.0936 -0.0465  0.1760
USO  0.1058 -0.1417  9.9429
XLY -0.0272 -0.1046 -0.2319
XLP  0.0687 -0.0189  0.2001
XLE -0.0440 -0.0361  2.0778
XLF  0.1168  0.0100 -0.2248
XLK -0.1167  0.0062 -0.0094
[1] "USO" "XLE" "XLP" "VNQ" "IEF" "XLK" "XLF" "XLY"
[1]  8 21
     2006  2007  2008  2009  2010  2011  2012  2013  2014  2015  2016  2017
[1,] "VNQ" "USO" "VNQ" "XLK" "IEF" "IEF" "IEF" "XLY" "VNQ" "XLY" "XLE" "IEF"
[2,] "XLP" "XLE" "IEF" "XLY" "XLY" "XLP" "XLF" "XLP" "IEF" "XLP" "XLF" "XLK"
[3,] "XLF" "IEF" "XLE" "XLP" "VNQ" "VNQ" "XLY" "XLK" "XLP" "XLK" "IEF" "XLP"
[4,] "IEF" "XLK" "XLP" "USO" "XLP" "XLY" "VNQ" "XLF" "XLK" "IEF" "XLK" "XLY"
[5,] "XLY" "XLP" "XLY" "IEF" "XLE" "XLE" "XLP" "IEF" "XLF" "VNQ" "VNQ" "VNQ"
[6,] "XLK" "XLY" "XLF" "XLE" "XLK" "XLK" "XLK" "XLE" "XLY" "XLF" "XLP" "USO"
[7,] "XLE" "VNQ" "XLK" "VNQ" "XLF" "USO" "XLE" "USO" "XLE" "XLE" "XLY" "XLF"
[8,] "USO" "XLF" "USO" "XLF" "USO" "XLF" "USO" "VNQ" "USO" "USO" "USO" "XLE"
     2018  2019  2020  2021  2022  2023  2024  2025  2026
[1,] "XLY" "IEF" "XLK" "USO" "XLE" "XLK" "XLF" "IEF" "USO"
[2,] "XLK" "VNQ" "IEF" "VNQ" "USO" "XLY" "USO" "XLF" "XLE"
[3,] "IEF" "XLP" "XLY" "XLE" "XLP" "IEF" "XLP" "XLK" "XLP"
[4,] "VNQ" "USO" "XLP" "XLF" "XLF" "XLF" "IEF" "XLP" "VNQ"
[5,] "XLP" "XLK" "XLF" "XLP" "XLK" "XLP" "XLY" "XLE" "IEF"
[6,] "XLF" "XLF" "VNQ" "XLK" "VNQ" "USO" "XLE" "VNQ" "XLK"
[7,] "XLE" "XLY" "XLE" "XLY" "IEF" "VNQ" "VNQ" "XLY" "XLF"
[8,] "USO" "XLE" "USO" "IEF" "XLY" "XLE" "XLK" "USO" "XLY"
```

E Independent verification record

```
PASS: Return matrices and named horizon outputs have the required shapes
PASS: Paired bootstrap returns equal sums of the underlying daily log returns
PASS: All five optimized weights agree with inverse-covariance calculations
PASS: Ten-year optimization agrees with the independent mathematical optimum
PASS: Risk function has the correct sign and handles arbitrary input data
PASS: CAPM tables and alpha matrix have eight assets and one column per interval
PASS: Every annualized alpha agrees with independent OLS regression
PASS: Every CAPM intercept and beta agrees with independent OLS regression
PASS: Each year's ranking contains every ETF once in descending alpha order
PASS: Both generated PNG plots exist and have the requested dimensions

All 10 independent checks passed.
Ten-year optimal VTI allocation: 0.4543241
Largest weight difference from analytic optimum: 1.62e-05
Largest annualized-alpha difference before rounding: 4.98e-05

Package warnings during execution:
M-step did NOT converge. Returning unconverged lM-estimate
```